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LLM-Generated Chart Explanations in Healthcare: A Computational and User-Centred Evaluation

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Abstract

Background. Healthcare data visualisations communicate essential information to diverse audiences, yet non-expert users frequently misinterpret charts, misread scales, and overlook key patterns. Large Language Models (LLMs) such as GPT-4o and Claude Sonnet 4.6 offer a scalable solution through automated natural language explanations, but critical uncertainties remain regarding their factual accuracy and whether their explanations genuinely support user comprehension in high-stakes healthcare contexts.

Aims. This study examined how accurately and effectively GPT-4o and Claude Sonnet 4.6 explain healthcare data visualisations, and whether explanation quality influences comprehension and confidence calibration among users with limited visualisation literacy.

Methods. Forty-five healthcare visualisations, stratified by chart type and complexity, were paired with expert-authored captions as ground-truth benchmarks. Ninety LLM explanations were evaluated using a 10-item accuracy checklist and a nine-type error taxonomy. A within-subjects user study (n = 40) compared comprehension, confidence, and confidence calibration across AI-assisted and chart-only conditions across six selected visualisations. Qualitative responses were analysed using reflexive thematic analysis.

Results. GPT-4o outperformed Claude Sonnet 4.6 on overall accuracy (median 8.00 vs 7.00, r = +0.75), with Claude producing unsupported inferential hallucinations on 43 of 45 explanations. Both models degraded monotonically with complexity. AI-assisted explanations were associated with substantially lower comprehension (Mdn = 7.00 vs 11.00, r = −0.72) and a markedly elevated false-confidence rate (51.7% vs 20.8%). Visualisation experience conferred no protective effect. Higher explanation accuracy correlated positively with comprehension (ρ = +0.83) but not with confidence. Users perceived explanations as helpful despite measurable comprehension decrements.

Conclusions. These findings caution against uncritical deployment of LLM-generated chart explanations in healthcare contexts. Despite being perceived as helpful, AI explanations impaired objective comprehension while sustaining high confidence - a miscalibration pattern that is particularly hazardous where accurate data interpretation carries real-world

consequences. AI explanation tools require calibration-aware design, including uncertainty signalling and prompted self-verification, before deployment in high-stakes contexts

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1. Introduction

1.1 Background Information

Healthcare data visualisations communicate essential information to diverse audiences, from epidemiological trends informing public health policy to treatment outcomes guiding patient decision-making (Abudiyab & Alanazi, 2022). However, presenting data visually does not guarantee understanding. Non-expert audiences often misinterpret charts, misread scales, or overlook key patterns, creating risks in high-stakes healthcare contexts (Turchioe et al., 2019).

Recent advances in multimodal Large Language Models (LLMs), particularly GPT-4o and Claude 4.6 Sonnet, present an automated alternative. These models can process chart images and generate natural language explanations without human intervention (OpenAI, 2024; Anthropic, 2024). The potential benefits are substantial: scalable accessibility for audiences with limited data literacy (Arcia, 2024); consistent explanation quality across large volumes of visualisations (Schulze et al., 2023); and rapid deployment during time-sensitive public health situations (Naumova, 2024). Unfortunately, critical uncertainties arise concerning the system's reliability in supporting healthcare data communication .

Two fundamental questions must be answered before deployment. First, can LLMs generate factually accurate explanations across different visualisation types and complexity levels, or do they exhibit systematic errors or biased trends that undermine quality (Wang et al., 2024)? Second, even if computationally accurate, do these explanations actually improve user comprehension compared to viewing visualisations alone ? Moreover, there is concern that

explanations that meet automated evaluation metrics may still mislead users or foster inappropriate confidence in faulty interpretations (De Bona et al., 2024).

This project addresses these uncertainties through systematic evaluation combining computational accuracy assessment with controlled user validation. By examining both (1) whether LLM explanations are accurate (see section 1.2) and (2) whether they genuinely help diverse users understand healthcare visualisations, this research provides evidence to inform responsible deployment decisions.

1.2 Aims and Objectives

In this study, **accuracy** refers to the factual correctness of LLM-generated explanations when verified against expert-authored captions, operationalised through systematic error annotation (Thomson & Reiter, 2020) and a typology adapted from Huang et al. (2024). **Effectiveness** encompasses both computational quality (readability, completeness, and error prevalence) and user-centred outcomes (comprehension improvement and perceived helpfulness), reflecting whether explanations fulfil their communicative purpose (Lundgard & Satyanarayan, 2022). **Confidence calibration** denotes the correspondence between participants' subjective certainty and their actual comprehension accuracy (Lin & Zabrucky, 1998), where miscalibration is particularly consequential in healthcare contexts.

Primary Aim:

To examine how accurately and effectively Large Language Models generate explanations for healthcare data visualisations, and to understand how explanation quality influences comprehension and confidence calibration among users with limited visualisation literacy.

Objectives:

1. **Curate 45 healthcare visualisations** (bar charts, line graphs, scatter plots) stratified by complexity and paired with expert captions as ground-truth benchmarks.
2. **Generate 90 LLM explanations** using one comprehensive prompt and evaluate them through factual-accuracy scoring, an error taxonomy, and readability analysis.
3. **Conduct a controlled within-subjects user study (n = 40) in a form of questionnaire** comparing comprehension, confidence, and confidence calibration with an LLM explanation versus without any explanation
4. **Assess whether frequency of visualisation use** is associated with user comprehension and confidence when interacting with LLM-generated explanations.
5. **Integrate computational and user findings** to examine whether explanation accuracy affects user comprehension and confidence at the chart level.
6. **Develop a digital artefact (Streamlit app)** presenting visualisations, LLM explanations, accuracy metrics, and deployment guidelines.

2. Literature Review

2.1 Comprehension Challenges in Visualisations

Decades of research demonstrate that visualisation interpretation poses significant cognitive challenges across populations and chart types. Even simple bar and line charts produce systematic errors among nationally representative adults (Galesic and Garcia-Retamero, 2011), while more demanding visual statistics, such as confidence intervals and error bars, are misread even by trained researchers (Belia et al., 2005). These difficulties are not random: they trace to consistent process-level barriers in encoding choice, channel decoding, and translating questions into visual queries (Kim and Heer, 2018; Nobre et al., 2024). Together, these findings establish the empirical baseline against which any explanatory intervention must be assessed.

2.2 Automated Chart Explanation and Multimodal LLMs

Automated chart understanding has evolved from template-based captioning systems (Demir et al., 2012) to neural approaches capable of generating fluent, data-driven descriptions (Obeid and Hoque, 2020; Kantharaj et al., 2022). Parallel work in chart question-answering highlights persistent challenges, with models often misreading values or inferring unsupported relationships (Masry et al., 2022). The emergence of multimodal LLMs such as GPT-4o and Claude has expanded these capabilities, enabling image-conditioned explanation generation. However, evaluations show mixed performance: models adequately handle simple charts yet struggle with multi-series figures, numerical precision, and value hallucination (Masry et al., 2022; Liu et al., 2023).

Bendeck and Stasko (2025) evaluated GPT-4 on a suite of visualisation literacy tasks and found that the model recognises trends and extrema reliably but struggles with simple value retrieval when the original dataset is unavailable, lacks the ability to reliably distinguish between colours, and occasionally suffers from hallucination and inconsistency. Lo and Qu (2025) extend this picture by showing that multimodal LLMs can detect misleading visualisations with moderate accuracy under Chain-of-Thought prompting, but performance varies substantially across misleader types and prompting strategies, suggesting that fluent output can mask substantive limitations. These error patterns are grounded in the broader NLG literature by Ji et al. (2023), whose distinction between intrinsic hallucination (output that contradicts the source) and extrinsic hallucination (output that cannot be verified from the source) provides the standard typology for classifying generation errors. Despite the rapid release of competing frontier models, no published work has directly compared the explanation quality of GPT-4o and Claude on healthcare visualisations, leaving practitioners without empirical guidance on model selection.

2.3 Evaluating Factual Accuracy in Chart Explanations

A core obstacle to linking the computational and behavioural strands is the question of how explanation accuracy should be measured. Traditional NLG metrics such as BLEU rely on surface-level n-gram overlap with reference outputs and correlate poorly with human judgments (Reiter, 2018), and they are not designed to detect factual errors or hallucinations in generated text (Thomson and Reiter, 2020). In data-to-text generation more broadly, accuracy has been operationalised as factual correctness: the degree to which generated statements faithfully represent the source data (Thomson and Reiter, 2020). Furthermore, Huang et al. (2024) adapt this principle to the chart domain through the CHOCOLATE taxonomy, a framework for annotating factual errors in chart captions, including value, label, trend, and hallucination errors. Such error-level annotation enables granular diagnosis of

model weaknesses but has been validated only on general-purpose chart corpora rather than domain-specific visualisations such as those used in healthcare communication, where factual fidelity carries particular weight.

2.4 User-Facing Effects: Comprehension, Confidence, and Over-Trust

A separate but related strand of work asks not whether AI explanations are accurate but what they do to the people who read them. Across domains, evidence shows that AI explanations can shape user behavior in ways that diverge from their apparent quality. Users frequently accept AI suggestions without analytical engagement, and explanations often deepen this reliance rather than prompting scrutiny (Buçinca et al., 2021). A particularly persistent finding is that confidence and accuracy are dissociable: Zhang et al. (2020) showed that displayed confidence scores can calibrate subjective trust without producing corresponding gains in decision accuracy, suggesting that feelings of certainty and actual understanding move independently. This dissociation becomes acutely consequential in healthcare, where misplaced confidence can translate into poor self-management or clinical judgment (Lin and Zabrucky, 1998; Hattie, 2013).

The same pattern emerges in clinical AI: Kayser et al. (2024), in a study of 85 practitioners interpreting chest X-rays, found that natural-language explanations produced significant over-reliance, with factually rich but misaligned explanations producing the worst outcomes. Vasconcelos et al. (2023) trace this pattern to a cost-benefit trade-off: verifying an AI's explanation is cognitively effortful, and when that effort is high, users default to agreement rather than checking. This effect may be amplified for LLM-generated chart explanations, whose surface fluency can itself bias judgments of credibility (Reber et al., 2004), leaving the

question of how such explanations affect confidence-accuracy calibration is largely unexamined in the current literature.

2.5 Research Gaps and Project Contribution

Three critical gaps emerge: no systematic evaluation of state-of-the-art LLM explanation quality across visualisation types and complexities using error-level annotation; no controlled study linking computational accuracy to user comprehension and confidence calibration; and no qualitative account of which explanation features users themselves perceive as helpful or unhelpful in healthcare contexts. This project addresses these gaps through mixed-methods integration, combining a binary error taxonomy with a within-subjects user comparison and thematic analysis of open-ended responses in healthcare contexts.

2.6 Research Questions

Main Research Question:

How accurately and effectively do GPT-4o and Claude 4.6 Sonnet explain healthcare data visualisations?

Sub-Research Questions:

RQ1: How do GPT-4o and Claude 4.6 Sonnet compare in explanation accuracy and error patterns across visualisation types and complexity levels?

RQ2a: How does the presence of AI-generated explanations affect users' comprehension, confidence, and confidence calibration when interpreting healthcare data visualisations?

RQ2b: What is the correlation between frequency of visualisation use and (i) user comprehension (ii) confidence when interacting with LLM-generated explanations?

RQ3: What is the correlation between the model's computational accuracy and (i) user comprehension and (ii) confidence when interacting with LLM-generated explanations?

RQ4: What features of AI-generated explanations do users perceive as helpful or unhelpful when interpreting healthcare data visualisations?

3. Methodology

This study employed a mixed-methods design combining a computational accuracy evaluation with a within-subjects user study, following established chart-captioning evaluation frameworks (Tang et al., 2023; Kantharaj et al., 2022). The approach was chosen because the research questions required both an objective measurement of explanation accuracy and a behavioural assessment of how those explanations affected users. A within-subjects design, in which each participant is exposed to every condition and serves as their own control, was selected for its statistical power advantage with modest sample sizes (Field, 2013).

3.1 Data Collection Procedure

Phase 1: Visualisation Corpus and LLM Generation

Forty-five healthcare visualisations were curated from open-access sources (PLOS ONE, Our World in Data, ONS/GOV.UK, WHO, BMJ Open, NHS Digital), stratified by chart type (15 bar, 15 line, 15 scatter) and complexity (15 simple, 15 moderate, 15 complex). Complexity was determined from structural features such as the number of data series and encoding channels, following Bertin (1983) and Munzner (2014) (View complexity framework in Appenidx G). Each visualisation was paired with its original expert-authored caption as the primary ground-truth benchmark, supplemented by direct visual inspection and second-rater verification (Section 3.2).

Explanations were generated through the GPT-4o and Claude Sonnet 4.6 APIs, producing 90 explanations (45 charts × 2 models). The prompt was developed de novo as a structured six-component instruction grounded in chart-to-text and health-communication literature (Lundgard & Satyanarayan, 2022; Lee et al., 2017; Okan et al., 2012), requiring the model to

identify the chart type, extract key values, describe trends, and provide an overall interpretation. The design followed prompt-engineering best practices including explicit role assignment, structured output specification, and stepwise instruction sequencing (Sahoo et al., 2024; Sivarajkumar et al., 2024), since structured prompts enhance interpretive accuracy by guiding model reasoning (Li et al., 2024). A 100 to 150 word constraint prevented accuracy inflation through hedging, the tendency of LLMs to introduce vague qualifications that obscure factual errors (Huang et al., 2024). A pilot test verified prompt consistency prior to full generation.

Phase 2: User Study

Forty non-expert participants (aged 18 or over, English-fluent) were recruited via university networks. Six visualisations were selected after Phase 1 coding to maximise checklist-score spread, supporting the variance required for RQ3. Two 20-minute counterbalanced Google Form versions (A and B) controlled for order effects (Field, 2013) (View Appendix B). In Form A, charts 1, 3 and 5 were presented without explanations and charts 2, 4 and 6 with explanations; in Form B, this order was reversed. This ensured each chart was viewed by half of the participants in each condition, so that effects of chart difficulty and learning could not be confounded with the AI condition. Low-accuracy charts containing visible LLM errors functioned as embedded trick items, allowing the analysis to test whether participants detected factual errors or accepted them uncritically.

For each visualisation, participants answered three multiple-choice questions targeting retrieval, trend interpretation, and comparison. The cognitive levels were grounded in Masry et al.'s (2022) chart-question-answering task types, which distinguish information lookup from higher-order reasoning, and the question structure drew on Boy et al. (2014), whose visualisation-literacy assessment validated similar formats for measuring comprehension. Distractors were constructed from plausible chart misinterpretations following Haladyna et

al. (2002), so that incorrect options reflected realistic reasoning errors rather than implausible guesses. After each chart, participants rated their confidence on a 1 to 5 Likert scale, an instrument shown to reliably capture metacognitive judgement when paired with binary accuracy outcomes (Fischhoff et al., 1977; Galesic & Garcia-Retamero, 2011). A pre-study section collected demographics and frequency of visualisation use (Never-Daily), and a post-study open-ended question asked what made the explanations helpful or unhelpful.

The survey was piloted in early March 2026 with four participants (10% of the target sample), consistent with recommended pilot proportions (Connelly, 2008). The pilot identified ambiguous wording in two comprehension questions and minor layout issues; the instrument was revised accordingly, and pilot responses were excluded from the main analysis.

3.2 Analysis Methods

All analyses were conducted in Python with a significance threshold of $\alpha = .05$. Effect sizes (Cohen's d or d_z, η^2, r) were reported for all tests. Inter-rater reliability for Phase 1 coding was assessed by an independent second coder who scored a 20% random subsample using the same checklist and error taxonomy, following standard practice for content coding (Hallgren, 2012). Cohen's κ was calculated per item and per error code, with disagreements resolved through discussion.

Two coding instruments were applied to each of the 90 explanations. The first was a 10-item binary checklist (C1 to C10) yielding a 0 to 10 accuracy score, with an accuracy subscale (C1 to C5, C8) at semantic Levels 1 and 2 and a comprehension subscale (C6, C7, C9, C10) at Levels 3 and 4 (Lundgard & Satyanarayan, 2022); a 5% numerical tolerance was applied to value-based items. The second was a 9-type binary error taxonomy adapted from the CHOCOLATE framework (Huang et al., 2024), comprising five factual-error codes (FE1 to

FE5), two hallucination codes (HM1 and HM2), and one critical-omission code (OE1). View Appendix F to view when each one fires.

Table 1 maps each research question to its statistical test, data type, and effect size. Non-parametric tests (Wilcoxon signed-rank, McNemar, Kruskal-Wallis) were selected throughout because the binary or ordinal nature of most outcomes, combined with the modest sample size, violated the normality assumptions of parametric alternatives (Field, 2013). The Benjamini-Hochberg false-discovery-rate correction was applied within the RQ1 checklist-item family (10 tests) and error-code family (9 tests) to control for multiple comparisons.

Table 1

Mapping of research questions to statistical tests, data types, and effect sizes.

RQ	Analysis (data type)	Test	Effect size
RQ1	Total accuracy, GPT-4o vs Claude (paired ordinal)	Wilcoxon signed-rank	Rank-biserial r
	Checklist items C1–C10 (paired binary)	McNemar; BH-FDR	Odds ratio
	Error codes FE1–OE1 (paired binary)	McNemar; BH-FDR	Odds ratio
	Accuracy by chart type (independent ordinal)	Kruskal-Wallis; Dunn post-hoc	ε^2
	Accuracy by complexity (ordered)	Kruskal-Wallis + Spearman	ε^2 and ρ
	Error distribution × type/complexity (categorical)	Chi-square (Fisher if cells < 5)	Cramér's V
RQ2a	Comprehension AI vs no-AI (paired ordinal)	Wilcoxon signed-rank	Rank-biserial r

	Confidence AI vs no-AI (paired ordinal)	Wilcoxon signed-rank	Rank-biserial r
	False-confidence count AI vs no-AI (paired count)	Wilcoxon signed-rank	Rank-biserial r
RQ2b	Frequency × comprehension (ordinal pair)	Spearman correlation	ρ
	Frequency × confidence (ordinal pair)	Spearman correlation	ρ
	Frequency × error-aligned count (exploratory)	Spearman correlation	ρ
RQ3	Chart accuracy × user outcomes, n = 6 charts (ordinal pair)	Descriptive Spearman only; no inferential p-value	ρ
RQ4	Helpful/unhelpful open-ended responses (qualitative)	Inductive thematic analysis (Braun & Clarke, 2006)	Theme frequency

Note: Benjamini-Hochberg FDR correction was applied within the RQ1 checklist-item family (10 tests) and error-code family (9 tests). The RQ2b error-alignment correlation is reported as exploratory given the narrow range of the dependent variable (0–4). RQ3 correlations are reported descriptively, without inferential interpretation, given that n = 6 charts.

Open-ended RQ4 responses were analysed using inductive thematic analysis following Braun and Clarke's (2006) six-phase procedure: familiarisation, initial coding, theme generation, theme review, defining and naming themes, and reporting with examples. An inductive approach was selected because no prior taxonomy of perceived helpfulness for LLM-generated chart explanations exists, and themes were therefore derived bottom-up from participants' own language.

3.3 Digital Artefact Construction

The digital artefact is an interactive Streamlit web application built with Streamlit, pandas, matplotlib, and Plotly, hosted on Streamlit Community Cloud. It contains four sections: a Visualisation Explorer (charts with captions, explanations, and accuracy data); a Model Evaluation Dashboard (scores, error distributions); Findings and Guidance; and Common Pitfalls.

Link: https://blank-app-rwv9plj8dn.streamlit.app/

3.4 Ethical Issues

This research complied with the University of Sheffield Research Ethics Policy, the British Psychological Society Code of Human Research Ethics, and the GDPR. Visualisations were sourced under Creative Commons licences. The study involved no deception or sensitive content and excluded minors. Participants received an information sheet and provided informed consent. Data were anonymised immediately, with no identifying information collected, and participants could withdraw without penalty. Data were stored securely with a 1-year retention period and then deleted. Ethics approval was obtained via the en bloc route, and all identified LLM errors, including medical misinformation, were reported transparently in the results.

4. Results

4.1 RQ1: Computational Accuracy and Error Patterns

4.1.1 Worked Example of Checklist Coding

To illustrate the coding procedure before reporting aggregate results, one worked example is presented. The example shows a moderate-complexity bar chart, the expert-authored caption taken from the original publication, and one LLM-generated explanation produced under the structured six-component prompt described in Section 3.2

Table 2

Chart, expert-authored ground truth, and Claude-generated takeaway

Field	Content

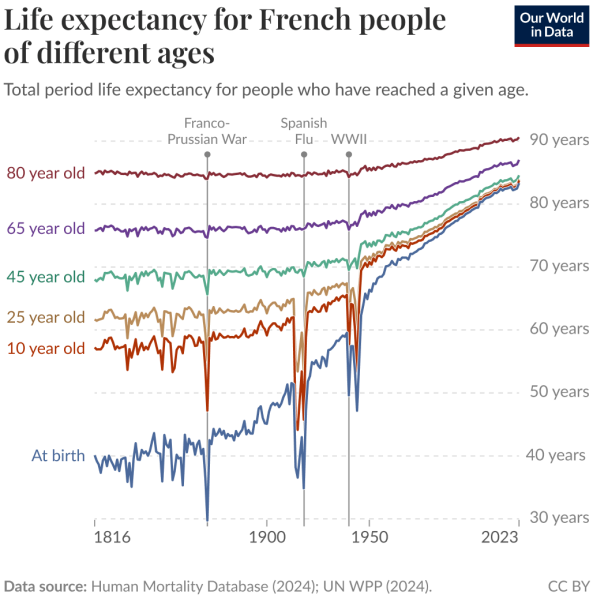
Chart	Life expectancy for French people of different ages, 1816–2023 
Source	Our World in Data, period life expectancy by age cohort
Expert-authored caption (ground truth)	Expert authored caption and Claude Sonnet 4.6 takeaway in Appendix C Table 1
Claude Sonnet 4.6 takeaway	Expert authored caption and Claude Sonnet 4.6 takeaway in Appendix C Table 1

Table 3

Checklist (C1-10) evaluation of Claude's takeaway

Code	Item	Result	Flagging rationale

C1	Chart type identified	**Pass**	Correctly identified in Claude takeaway section 1
C2	X-axis identified	**Pass**	Correctly stated in Claude takeaway section 2
C3	Y-axis identified	**Pass**	Correctly stated in Claude takeaway section 2
C4	Highest value correct	**Pass**	Matches paper's caption
C5	Lowest value correct	**Fail**	Value 30: correct, Duration: wrong " during the Spanish War"
C6	Direction correct	**Pass**	Correctly stated in Claude takeaway section 3
C7	Pattern correct	**Pass**	Correctly stated in Claude takeaway section 3
C8	Comparison correct	**Pass**	Correctly stated in Claude takeaway section 5
C9	Takeaway free of fabrication	**Fail**	Introduces external causal mechanisms (modern medicine, infectious disease management, age-related conditions management) not present in the chart's data
C10	Captures paper's primary finding	**Fail**	Directly inverts paper's primary point: paper says older-age improvements have been *more* important in recent decades; Claude says gains were "driven predominantly by reducing early-life mortality *rather than* extending old age"
	Checklist total	**7/10**	

Table 4

Error taxonomy coding (FE1–FE5, HM1, HM2, OE1) of Claude's takeaway

Code	Item	Result	Flagging rationale
FE1	Wrong value	**Fired**	Claude: By 2023, all age groups converge more closely (~78–90 years); Actual is 82-90
FE2	Axis/label error	**Fired**	Claude stated “Spanish Flu” instead of “Franco-Prussian War”
FE3a	Wrong direction	**Not fired**	Correctly stated
FE3b	Wrong pattern characterisation	**Not fired**	Correctly stated
FE4	Wrong magnitude descriptor	**Fired**	"Only modestly" and "incremental gains" understate an 8-year increase concentrated in the post-1950 period, which the paper emphasises as the chart's most important feature
FE5	Wrong entity at extremum	**Fired**	Lowest value’s y-axis misidentified as seen in C5 of Checklist
HM1	Fabricated factual content	**Not fired**	No fabricated content
HM2	Causal/policy/clinical mechanism	**Fired**	Introduces external causal mechanisms (modern medicine, infectious disease management, age-related conditions management) not present in the chart's data
OE1	Omission of paper's primary finding	**Fired**	Inverts paper's primary finding: “older-age improvements have been *more* important in recent decade”s; Claude says gains were "driven predominantly by reducing early-life mortality *rather than* extending old age"
	Errors fired	**6**	

4.1.2 Overall Model Comparison

Across the 45 paired charts, GPT-4o achieved a median checklist score of 8.00 (IQR [6.00, 9.00]) compared with 7.00 (IQR [5.00, 8.00]) for Claude Sonnet 4.6. A Wilcoxon signed-rank test on the 45 paired scores indicated a significant advantage for GPT-4o, $W = 61.50$, $p < .001$, $r = +0.75$, a large matched-pairs rank-biserial effect. Of the 45 paired comparisons, GPT-4o scored higher on 26 charts, Claude scored higher on 5, and 14 were tied. Subgroup descriptives by chart type and complexity band are presented in Table 5 and visualised in Figure 1 & 2.

Table 5

Checklist accuracy (out of 10) per chart by model, overall and by chart type and complexity

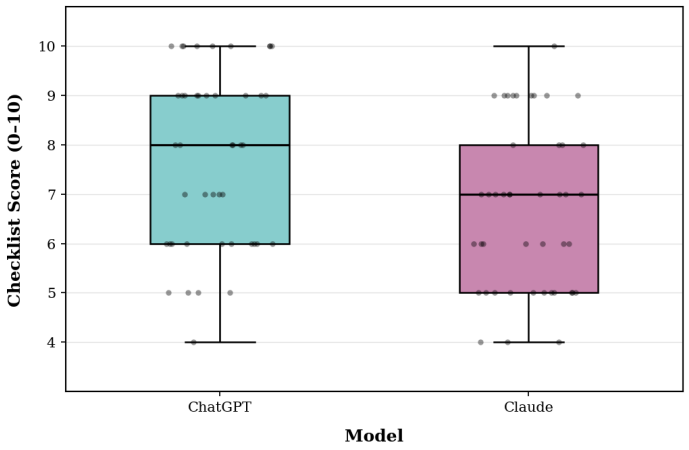
Model	Subgroup	n	Median	Mean (SD)	IQR	Min–Max
GPT-4o	Overall	45	8.00	7.71 (1.78)	[6.00, 9.00]	4–10
GPT-4o	Bar	15	9.00	8.00 (1.81)	[6.00, 9.50]	5–10
GPT-4o	Line	15	8.00	7.67 (1.63)	[6.00, 9.00]	5–10
GPT-4o	Scatter	15	8.00	7.47 (1.96)	[6.00, 9.00]	4–10
GPT-4o	Simple	15	9.00	9.07 (1.22)	[9.00, 10.00]	6–10
GPT-4o	Moderate	15	8.00	7.87 (1.46)	[7.00, 9.00]	5–10
GPT-4o	Complex	15	6.00	6.20 (1.37)	[5.50, 6.50]	4–9
Claude	Overall	45	7.00	6.71 (1.67)	[5.00, 8.00]	4–10
Claude	Bar	15	7.00	7.20 (1.70)	[5.50, 9.00]	5–9
Claude	Line	15	7.00	6.93 (1.53)	[6.00, 8.00]	4–9
Claude	Scatter	15	6.00	6.00 (1.65)	[5.00, 6.50]	4–10
Claude	Simple	15	9.00	8.20 (1.37)	[7.50, 9.00]	5–10
Claude	Moderate	15	7.00	6.33 (1.50)	[5.00, 7.00]	4–9

Claude	Complex	15	5.00	5.60 (0.91)	[5.00, 6.00]	4–7

Note. Scores are the count of the 10 binary checklist criteria met by each explanation. IQR is the interquartile range.

Figure 1

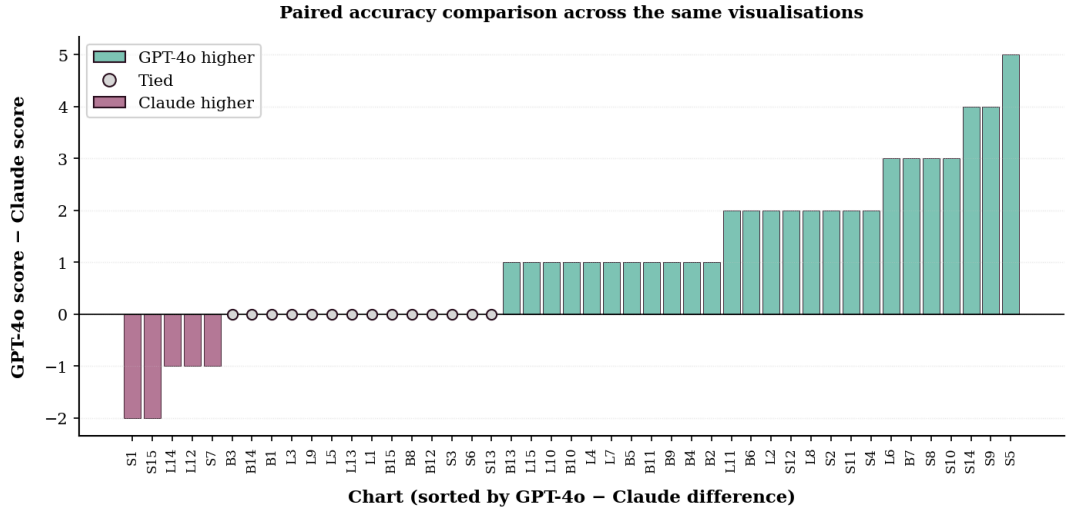
Distribution of total checklist scores by model



Note. Boxes show the median and interquartile range across the 45 charts per model; whiskers extend to 1.5 × IQR; semi-transparent points show individual chart scores.

Figure 2

Within-chart accuracy difference (GPT-4o − Claude) across the 45 visualisations



Note. Each bar represents one chart, sorted from largest Claude advantage on the left to largest GPT-4o advantage on the right. The largest GPT-4o advantage on a single chart was 5 points; the largest Claude advantage was 2 points.

4.1.3 Item-Level Checklist Performance

Per-item performance for the 10 binary checklist criteria was compared between models using paired McNemar tests (asymptotic with continuity correction; Fisher's exact two-sided test reported as a supplementary p-value when any cell n < 5), with Benjamini-Hochberg FDR correction across the 10 items as the primary correction. Two items showed a statistically reliable model difference after BH correction, both favouring GPT-4o: C5 (lowest value: 36 vs 26 passes; OR = 7.67, 95% CI [1.40, 42.03], p_BH = .047) and C9 (absence of hallucination: 23 vs 2 passes; OR = 9.40, 95% CI [2.55, 34.63], p_BH = .001). Under the more conservative Holm-Bonferroni correction, only C9 retained significance (p_Holm = .001). C1 (chart type) was passed by both models on all 45 charts. C7 (trend pattern) and C10 (overall takeaway) had the lowest pass counts for both models (≤ 23 of 45 each). Item-level results are reported in Table 6 and visualised in Figures 3 and 4.

Table 6

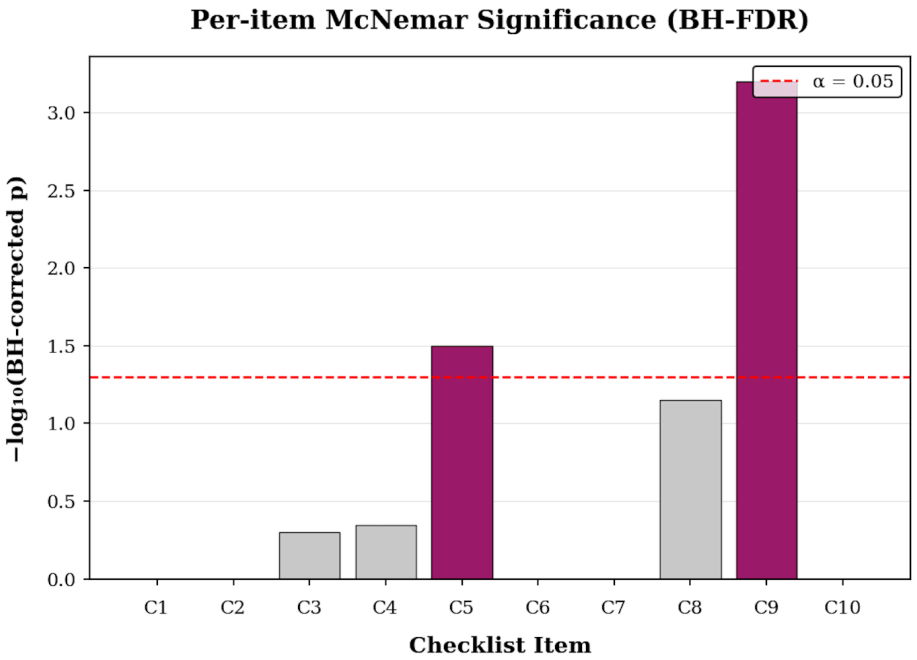
Per-item checklist results by model with paired McNemar comparison (n = 45 charts)

Item	Description	GPT pass	Claude pass	GPT only	Claude only	OR [95% CI]	*p* (BH)	Sig.
C1	Chart type	45	45	0	0	—	1.000	
C2	X-axis	44	44	1	1	1.00 [0.10, 9.61]	0.799	
C3	Y-axis	41	44	0	3	0.14 [0.01, 2.77]	0.496	
C4	Highest value	39	34	7	2	3.00 [0.72, 12.55]	0.456	
C5	Lowest value	36	26	11	1	7.67 [1.40, 42.03]	0.047	✓
C6	Trend direction	39	37	4	2	1.80 [0.38, 8.45]	0.976	
C7	Trend pattern	19	18	5	4	1.22 [0.35, 4.25]	1.000	
C8	Comparison	39	29	13	3	3.86 [1.19, 12.50]	0.081	
C9	No hallucination	23	2	23	2	9.40 [2.55, 34.63]	0.001	✓
C10	Overall takeaway	22	23	2	3	0.71 [0.14, 3.62]	1.000	

Note: GPT-only and Claude-only are the discordant cells (charts on which only that model passed). ORs are GPT-4o versus Claude; values > 1 indicate a higher pass rate for GPT-4o. Primary *p*-values are Benjamini-Hochberg FDR-adjusted across the 10 items. Items significant after BH correction are marked ✓. Under Holm-Bonferroni correction, only C9 retained significance.

Figure 3

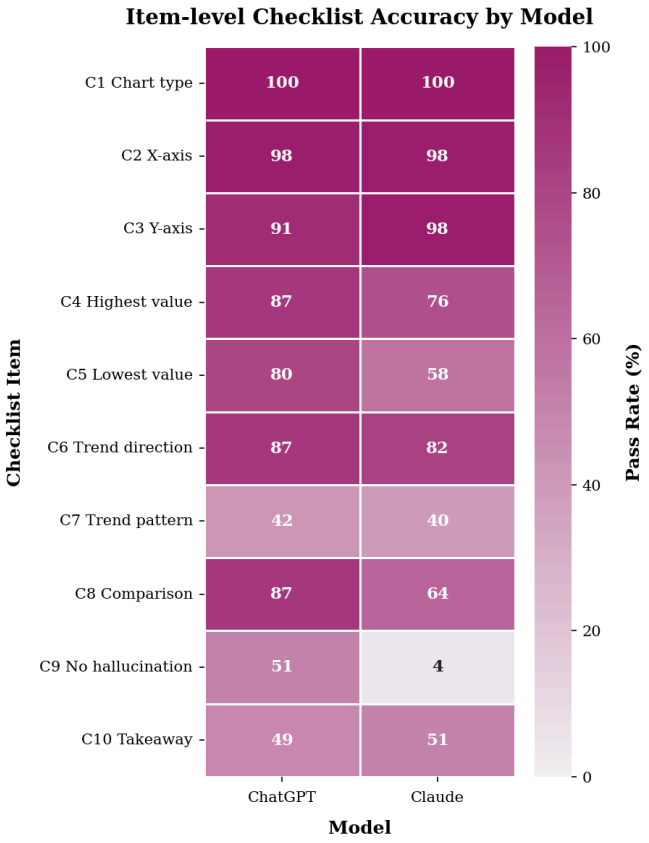
Per-item McNemar significance (BH-FDR adjusted) across the 10 checklist items



Note. Bars show $-\log_{10}(p_BH)$; the red dashed line marks the $\alpha = .05$ threshold ($-\log_{10}(.05) \approx 1.30$). Items above this line are significant after Benjamini-Hochberg correction (C5, C9).

Figure 4

Item-level checklist pass rate (%) per model



Note. Each cell shows the percentage of the 45 charts on which the explanation produced by that model satisfied the checklist criterion. Darker shading indicates higher pass rates.

4.1.4 Error Code Distribution

Across the 90 explanations, 9 error codes were applied (FE1–FE5, HM1–HM2, OE1). The same paired McNemar testing procedure with BH-FDR correction was used. Two codes differed significantly between models after BH correction, both more frequent for Claude than GPT-4o: FE5 (extremum errors: 13 vs 23 charts containing the error; OR = 0.13 in favour of GPT-4o, 95% CI [0.02, 0.72], p_BH = .042) and HM2 (content hallucination: 22 vs 43 charts; OR = 0.11, 95% CI [0.03, 0.39], p_BH = .001). Under Holm correction, only HM2 retained significance. FE3b (relationship errors) and HM2 were the most prevalent codes for both models, each appearing on at least 26 of 45 charts.

Table 7

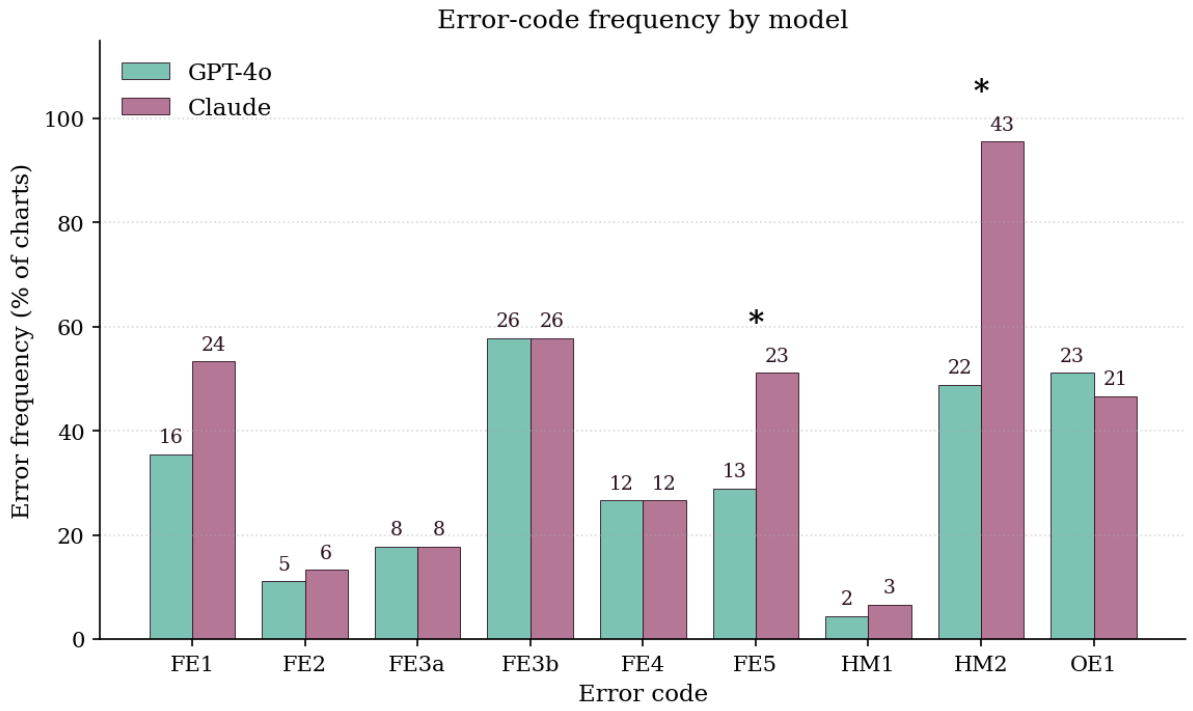
Per-code error results by model with paired McNemar comparison (n = 45 charts)

Code	Description	GPT (n)	Claude (n)	GPT only	Claude only	OR [95% CI]	*p* (BH)	Sig.
FE1	Incorrect value	16	24	4	12	0.36 [0.12, 1.06]	0.240	
FE2	Wrong label/axis	5	6	4	5	0.82 [0.24, 2.84]	1.000	
FE3a	Wrong trend direction	8	8	2	2	1.00 [0.17, 5.77]	1.000	
FE3b	Wrong trend pattern	26	26	4	4	1.00 [0.27, 3.69]	1.000	
FE4	Magnitude error	12	12	7	7	1.00 [0.36, 2.75]	1.000	
FE5	Extremum error	13	23	1	11	0.13 [0.02, 0.72]	0.042	✓
HM1	Fabricated data	2	3	2	3	0.71 [0.14, 3.62]	1.000	
HM2	Unsupported inference	22	43	2	23	0.11 [0.03, 0.39]	0.001	✓
OE1	Critical omission	23	21	4	2	1.80 [0.38, 8.45]	1.000	

Note. Columns follow the same format as Table 2. ORs < 1 indicate the code was more frequent for Claude than GPT-4o. p-values are BH-FDR-adjusted across the 9 codes. Codes significant after BH correction are marked ✓. Under Holm correction, only HM2 retained significance.

Figure 5

Error-code frequency by model



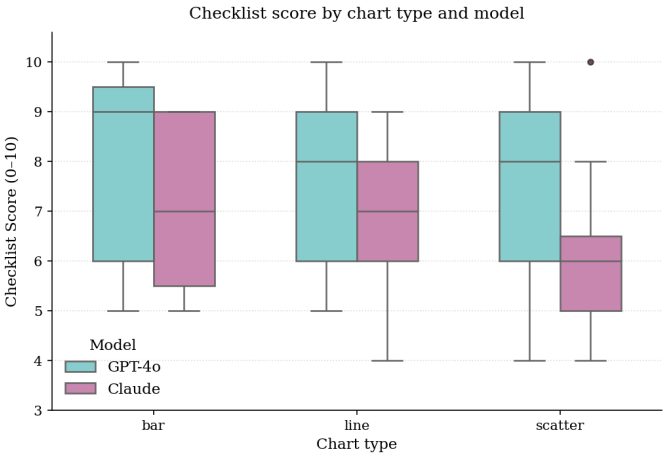
Note. Each pair of bars shows the error count of the 45 charts containing at least one instance of each error code. Asterisks (*) mark codes significant after BH-FDR correction (FE5, HM2).

4.1.5 Effects of Chart Type

Pooling across models, checklist score did not differ significantly across chart types (bar, line, scatter), $H(2) = 3.59$, $p = .166$, $\varepsilon^2 = 0.018$, a negligible effect. Dunn's post-hoc tests with Bonferroni correction confirmed that no pairwise comparison was significant: bar vs line $p = 1.000$, bar vs scatter $p = .190$, line vs scatter $p = .628$.

Figure 6

Checklist score by chart type and model



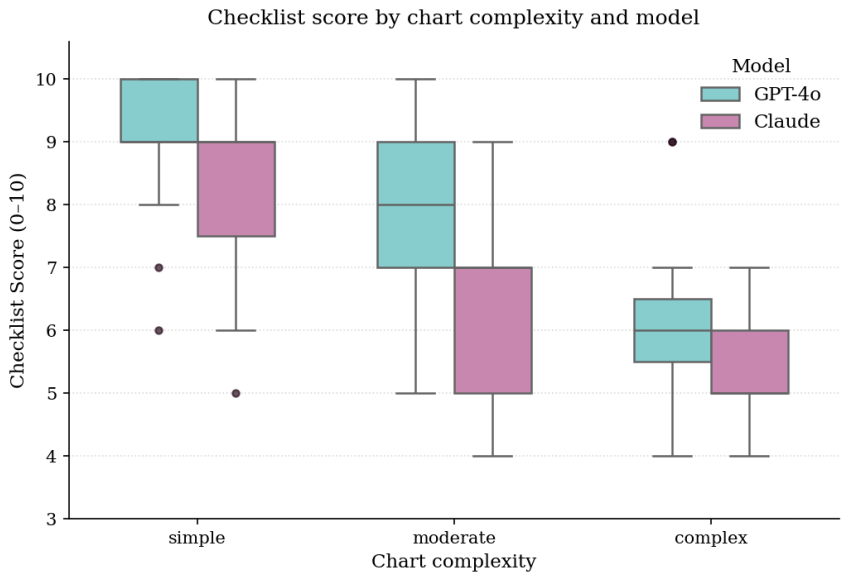
Note. Boxes show median and interquartile range across the 15 charts per cell; whiskers extend to 1.5 × IQR. Distributions overlap substantially across chart types; Kruskal-Wallis indicated no significant overall effect (p = .166).

4.1.6 Effects of Chart Complexity

Pooling across models, checklist score decreased monotonically with chart complexity (Spearman $\rho = -0.63$, $p < .001$, $n = 90$). A Kruskal-Wallis test confirmed a large effect across the three complexity bands, $H(2) = 35.21$, $p < .001$, $\varepsilon^2 = 0.38$. Because complexity is an ordered factor, Spearman ρ is the principal test of association; Dunn's pairwise post-hoc tests with Bonferroni correction are reported for completeness and showed significant differences between all three pairs: simple vs moderate $p = .003$, simple vs complex $p < .001$, moderate vs complex $p = .024$. The negative association held within each model (GPT-4o: $\rho = -0.67$, $p < .001$; Claude: $\rho = -0.63$, $p < .001$).

Figure 7

Checklist score by chart complexity and model



Note. Boxes show median and interquartile range across the 15 charts per cell. Both models decline steadily from simple to complex charts; Kruskal-Wallis indicated a large effect (ε^2 = 0.38), and Dunn's post-hoc with Bonferroni correction confirmed all three pairwise differences (simple vs moderate p = .003; simple vs complex p < .001; moderate vs complex p = .024).

4.1.7 Error Distribution by Chart Type and Complexity

To examine whether specific error codes were systematically associated with certain chart types or complexity bands, chi-square tests of independence (with Fisher's exact two-sided used in 2 × 2 cells where any expected count fell below 5) were applied to each error code crossed with chart type and complexity, with Cramér's V as the effect size. Results are shown in Table 4. Six error codes showed significant complexity associations after inspection of the raw p-values: FE1 (value errors), FE3a (comparison errors), FE3b (relationship errors), FE5 (extremum errors), HM2 (content hallucinations), and OE1 (essential omissions), with effect sizes ranging from small (HM2, V = 0.29) to large (OE1, V = 0.53). Each of these codes was more prevalent in moderate and complex charts than in simple ones. Chart-type associations

were considerably weaker, reaching the $p < .05$ threshold only for FE1 ($V = 0.39$) and FE3b ($V = 0.32$).

Table 8

Error code associations with chart type and complexity (χ^2 with Cramér's V)

Code	Chart-type χ^2	Chart-type *p*	Chart-type *V*	Complexity χ^2	Complexity *p*	Complexity *V*
FE1	13.59	.001	0.39	15.21	< .001	0.41
FE2	0.83	.661	0.10	5.80	.055	0.25
FE3a	0.15	.927	0.04	17.03	< .001	0.44
FE3b	9.38	.009	0.32	24.69	< .001	0.52
FE4	3.07	.216	0.19	4.43	.109	0.22
FE5	5.83	.054	0.26	14.44	.001	0.40
HM1	0.42	.809	0.07	0.42	.809	0.07
HM2	0.78	.679	0.09	7.42	.024	0.29
OE1	0.62	.733	0.08	25.17	< .001	0.53

Note. Chi-square test of independence between each error code and the factor; Fisher's exact two-sided was used where any expected cell count fell below 5 in a 2 × 2 table. Cramér's V is the effect size (small: 0.10, medium: 0.30, large: 0.50). p-values are uncorrected within each factor.

4.2 RQ2a: User Comprehension and Confidence

Forty participants completed the within-subjects study, each viewing six charts (three with an AI explanation, three without). For each chart, participants answered three weighted multiple-choice questions (1, 2, and 3 points; maximum 6 per chart, 18 per condition) and rated their confidence on a 1–5 scale (sum 3–15 per condition). Counterbalancing across the two groups ensured that each chart appeared in the AI condition for one group and the chart-only condition for the other.

4.2.1 Comprehension

Comprehension was lower in the AI-assisted condition (*Mdn* = 7.00, IQR [4.00, 10.25]) than in the chart-only condition (*Mdn* = 11.00, IQR [7.75, 15.00]). A Wilcoxon signed-rank test on the 40 paired sums indicated a significant decrement with AI assistance, $W = 87.50$, $p < .001$, $r = -0.72$, a large matched-pairs rank-biserial effect (Figure 8).

4.2.2 Confidence (Raw Rating)

Confidence sums were higher in the AI condition (*Mdn* = 11.00) than in the chart-only condition (*Mdn* = 9.50), but the difference did not reach the conventional significance threshold, $W = 114.50$, $p = .071$, $r = +0.39$, a medium effect (Figure 8b). The directional pattern of higher confidence under AI despite lower accuracy is visible in the side-by-side comparison.

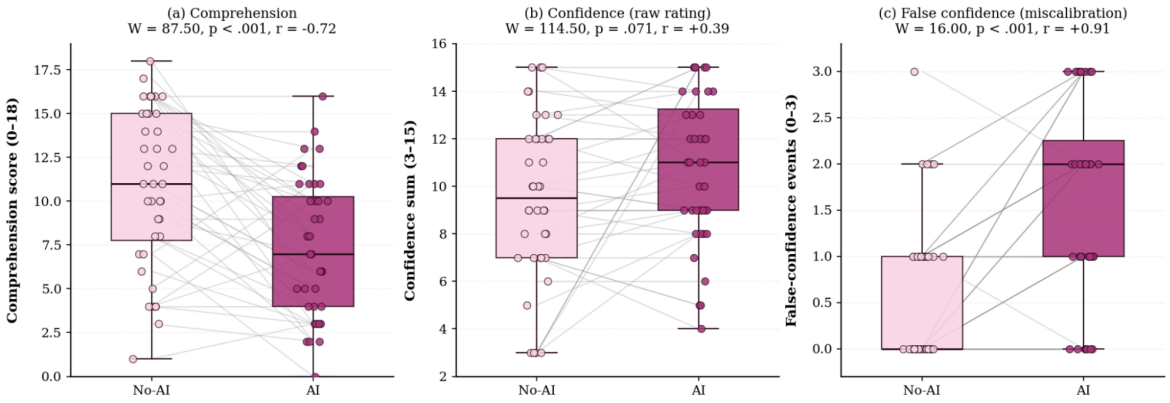
4.2.3 False Confidence (Calibration)

A chart response was classified as a false-confidence event when the participant rated their confidence at 4 or 5 of 5 but did not achieve the maximum 6/6 score on that chart's three questions, indicating a gap between subjective confidence and objective accuracy. At the per-response level, 62 of 120 AI-condition chart responses (51.7%) met this criterion,

compared with 25 of 120 chart-only responses (20.8%), a rate ratio of 2.48. At the participant level, the mean number of false-confidence events was 1.55 (out of 3) under AI versus 0.62 under chart-only. A Wilcoxon signed-rank test on the paired participant-level counts indicated a highly significant increase under AI, $W = 16.00$, $p < .001$, $r = +0.91$, a near-ceiling matched-pairs rank-biserial effect (Figure 8).

Figure 8

Within-subjects comparison of comprehension, confidence, and false-confidence counts under AI-assisted versus chart-only conditions (n = 40 participants)



Note. Each pair of points connected by a grey line represents one participant's scores under the two conditions. Box overlays show the median and interquartile range; whiskers extend to 1.5 × IQR. Panels show (a) comprehension score, (b) confidence rating, and (c) false-confidence count, all per participant per condition.

4.3 RQ2b: User Comprehension and Confidence

4.3.1 Exploratory: Individual Differences in Visualisation Use

Frequency of visualisation use was treated as an ordered scale (0 = Never, 1 = Rarely, 2 = Occasionally, 3 = Regularly, 4 = Daily) and correlated with AI-condition outcomes using

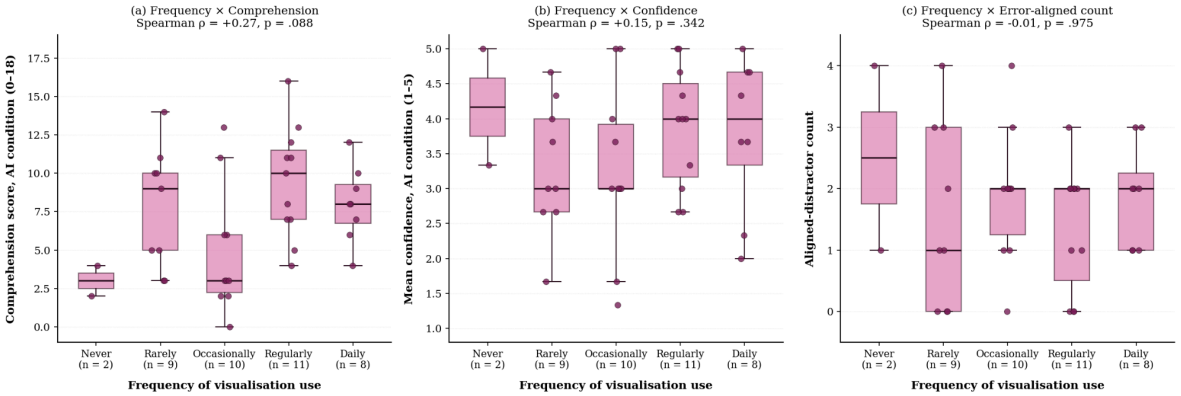
Spearman rank correlations. All 40 participants provided valid data; the distribution was 2 Never, 9 Rarely, 10 Occasionally, 11 Regularly, and 8 Daily.

4.3.2 Frequency and Comprehension

Frequency of use was positively but non-significantly associated with AI-condition comprehension score, ρ = +0.27, p = .088, 95% CI [−0.04, +0.54], n = 40 (Figure 9; 10). Group medians rose from 3.00 at Never to 10.00 at Regularly, but the trend was non-monotonic, with Occasionally users (median 3.00) sitting below Rarely users (median 9.00).

Figure 9

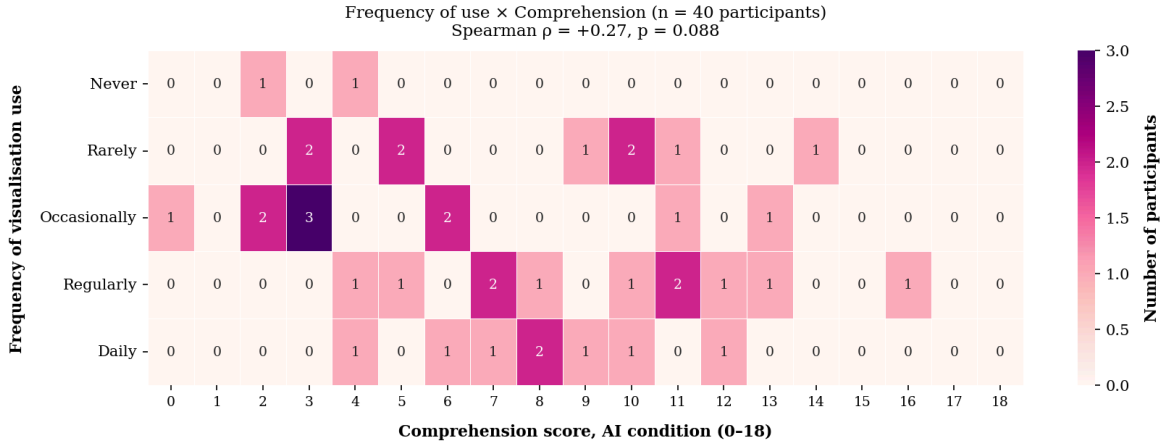
Distribution of AI-condition outcomes by frequency of visualisation use



Note. Per-participant outcomes across the five frequency levels (n = 40; counts shown in parentheses). Boxes show median and IQR; whiskers extend to 1.5 × IQR; circles are individual participants (jittered). (a) Comprehension score (0–18). (b) Mean confidence rating (1–5). (c) Aligned-distractor count. Spearman ρ above each panel; no correlation reached α = .05.

Figure 10

Joint distribution of frequency of use and AI-condition comprehension score



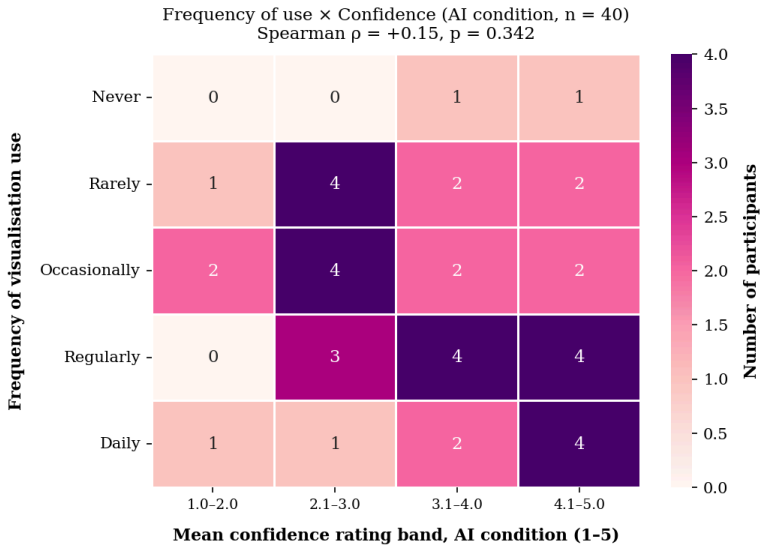
Note. Cells show participant counts (n = 40) at each combination of frequency level (rows; 0–4) and comprehension score (columns; 0–18). Darker shading = more participants. Spearman ρ = +0.27, p = .088.

4.3.3 Frequency and Confidence

Frequency of use was weakly and non-significantly associated with mean AI-condition confidence rating, ρ = +0.15, p = .342, 95% CI [−0.17, +0.44], n = 40 (Figure 9; Figure 11). Median ratings ranged narrowly from 3.00 to 4.17 across levels, with substantial spread at each.

Figure 11

Joint distribution of frequency of use and AI-condition mean confidence rating



Note. Cells show participant counts (n = 40) at each combination of frequency level (rows; 0–4) and mean confidence rating (columns; binned across 1–5). Darker shading = more participants. Spearman ρ = +0.15, p = .342.

4.3.4 Susceptibility to AI Errors

Responses on the four error-containing AI charts (140 chart-question responses) were classified as aligned (chose the AI-induced distractor), resisted (correct), or other (different wrong answer). Across all responses, 48.6% were aligned, 27.1% resisted, and 24.3% other; the AI-induced distractor was the modal response. Frequency of use was uncorrelated with per-participant aligned count, ρ = −0.005, p = .975, 95% CI [−0.32, +0.31], n = 40 (Figure 9c), indicating no protective effect of self-reported experience.

4.4 RQ3: Computational Accuracy and User Outcomes

RQ3 examines the chart-level relationship between an explanation's computational accuracy and user outcomes in the AI-assisted condition. Because the 6 user-study charts span a narrow accuracy range (4–9 out of 10) and the analysis unit is the chart (n = 6), Spearman rank correlations are reported descriptively. Chart-level summaries are shown in Table 9.

Table 9

Chart-level summary for the six user-study charts, AI condition (n=20)

Chart	Type	Complexity	Accuracy (/10)	Has Error	Mean Comp. (/6)	Mean Conf. (1–5)
1	Bar	Moderate	8	No	3.40	4.15
2	Line	Simple	9	No	2.85	3.30
3	Line	Moderate	7	Yes	3.00	4.00
4	Scatter	Moderate	4	Yes	0.95	3.70
5	Scatter	Complex	5	Yes	2.00	3.15
6	Bar	Complex	6	Yes	2.45	3.25

4.4.1 Accuracy and Comprehension

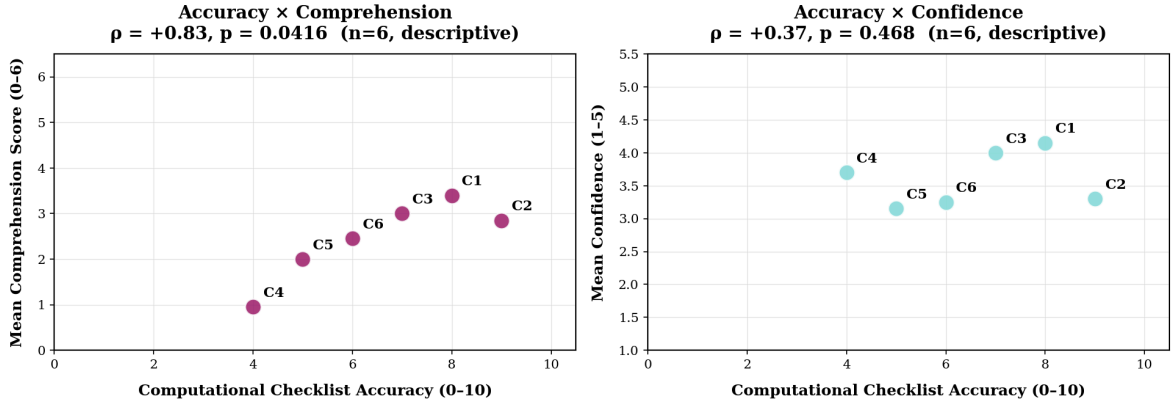
Across the six user-study charts, higher checklist accuracy was associated with higher mean participant comprehension on that chart, $\rho = +0.83$, $p = .042$, $n = 6$ (Figure 12). Chart 4 (accuracy 4/10) had the lowest mean comprehension (0.95/6); Chart 1 (accuracy 8/10, no error codes applied) had the highest (3.40/6).

4.4.2 Accuracy and Confidence

The chart-level association between checklist accuracy and mean confidence was small and not statistically significant, $\rho = +0.37$, $p = .468$, $n = 6$ (Figure 12). Confidence ratings remained in a narrow band (mean 3.15–4.15 across the six charts), with no monotonic relationship between explanation accuracy and reported confidence.

Figure 12

Chart-level relationship between computational accuracy and AI-condition user outcomes



Note. Each point represents one of the six user-study charts (n = 6 charts, 20 participants per chart). (a) Mean comprehension score plotted against computational checklist accuracy (b) Mean confidence rating (1–5) plotted against computational checklist accuracy. Chart labels (C1–C6) correspond to chart identifiers in Table 4. Spearman ρ and two-sided p shown above each panel. Because n = 6 charts, both correlations are reported descriptively.

4.5. RQ4: Qualitative Themes on Helpfulness

All 40 participants provided open-ended responses on what made the AI explanations helpful or unhelpful. Seven (17.5%) were excluded as off-topic essays on LLMs unrelated to the study, leaving 33 substantive responses (mean length 260 characters, median 154). Responses were analysed using reflexive thematic analysis (Braun and Clarke, 2006, 2019); the full procedure is described in Section 3.2. Each response could carry multiple theme tags.

Five themes were identified (Table 10).

Table 10

Themes identified in 33 substantive responses on the helpfulness of AI explanations

Theme	n	%	Definition
1. Aided comprehension and speed	25	75.8	Explanation oriented attention, surfaced key values, or accelerated answering.

2. Difficulty with healthcare content / complexity	4	12.1	Healthcare jargon or visual density (overlapping points, many variables) limited helpfulness.
3. Error awareness	4	12.1	Participant noticed factual inaccuracies in the explanation.
4. Insufficient depth or specificity	6	18.2	Explanation gave a general overview but did not address question specifics.
5. Resistance or preference for chart alone	6	18.2	Participant relied on the chart, ignored the explanation, or focused on chart features rather than the AI text.

Note. Frequencies are response-level counts; percentages are out of 33 substantive responses. Multi-tagging permitted.

Theme 1 was by far the most common, with participants describing the explanation as directing attention to key values or speeding up answering (Group2_Participant01: "The explanations gave me an idea on how to narrow down the options and answer better and direct my focus into certain values"). Theme 2 was raised by participants pointing to jargon or visually dense charts (G1_P05: "When there are so many variables or factor present and especially visually overlapping it is unclear"). Theme 3 was carried by three frequent or daily visualisation users, with two independently flagging the Scotland–England alcohol mortality comparison (Chart 5) as misleading (G2_P12; G2_P14). Theme 4 captured responses that valued the orientation function but found explanations too general (G1_P17: "The LLM explanation helped me answer the questions faster but they didn't help answer all the questions not addressing all the key points asked"). Theme 5 included participants who relied on the chart over the text (G2_P19: "I was focusing more on the chart rather than reading the explanation") and others whose responses described chart features (colour, axes, legend) rather than the AI text, suggesting limited engagement with the explanation as a separate object.

5. Discussion and Conclusion

5.1 Summary of Findings in Relation to Research Questions

RQ1: Comparative accuracy and error patterns. GPT-4o outperformed Claude Sonnet 4.6 on overall checklist accuracy (median 8.00 vs 7.00, r = +0.75), with Claude's disadvantage driven by more frequent extremum misidentification (FE5) and substantially higher rates of unsupported inferential content (HM2). Only lowest-value identification (C5) and absence of hallucination (C9) separated the models after BH correction. The error taxonomy confirmed this, with content hallucinations (HM2) appearing on 43 of 45 Claude explanations versus 22 for GPT-4o. Both models degraded monotonically with chart complexity ($\rho = -0.63$ to -0.67), while chart type had no significant effect, suggesting information density rather than visual encoding format is the primary driver of accuracy loss. These patterns align with Masry et al. (2022) and Bendeck and Stasko (2025), who found that multimodal LLMs handle trends adequately on simpler charts but struggle with numerical precision under greater visual complexity. Claude's hallucination profile is consistent with Ji et al.'s (2023) category of extrinsic hallucination - fluent content that cannot be verified from the source - manifesting here as introduced causal mechanisms absent from the underlying data.

RQ2a: Effects on comprehension, confidence, and calibration. AI-assisted explanations were associated with substantially lower comprehension (Mdn = 7.00 vs 11.00, r = −0.72) alongside a non-significant trend toward higher confidence. The false-confidence rate sharpens this picture: 51.7% of AI-conditioned responses were high-confidence errors versus 20.8% in the chart-only condition (r = +0.91). This directly mirrors Kayser et al.'s (2024) finding that natural-language clinical AI explanations produce over-reliance when fluent but misaligned and is consistent with Vasconcelos et al.'s (2023) cost-benefit account -

participants accepted the explanation as an interpretive shortcut rather than verifying it against the chart. Reber et al.'s (2004) processing fluency work is also relevant: surface polish may itself signal credibility, inflating confidence independently of accuracy.

RQ2b: Individual differences. Frequency of visualisation use was not significantly associated with comprehension ($\rho = +0.27$), confidence ($\rho = +0.15$), or susceptibility to AI-induced errors ($\rho = -0.005$). The near-zero error-susceptibility correlation is particularly notable: experience conferred no protective effect, with the AI-induced distractor, the modal response (48.6% aligned). This is consistent with Buçinca et al. (2021), who found that habitual AI engagement can deepen rather than moderate over-reliance.

RQ3: Computational accuracy and user outcomes. Higher explanation accuracy was positively associated with chart-level comprehension ($\rho = +0.83$, $p = .042$), suggesting that more factually accurate AI explanations were generally linked to better participant understanding. However, confidence remained relatively stable across accuracy levels ($\rho = +0.37$, ns), reinforcing a calibration dissociation: users felt similarly certain regardless of whether the explanation they read was accurate. This indicates a potential false-confidence effect, where fluent AI explanations may increase perceived understanding without reliably improving actual performance. This is consistent with Zhang et al. (2020), who found that confidence and accuracy can move independently, with subjective trust improving even when decision accuracy does not. The result should be interpreted cautiously given the small chart-level sample ($n = 6$).

RQ4: Perceived helpfulness. The dominant theme, aided comprehension and speed (75.8%), suggests that participants perceived AI explanations as useful orientation tools. However, this did not fully align with participant comprehension rates, indicating a possible

subjective–objective gap: users may have felt helped even when measurable understanding did not improve. This again points to a false-confidence risk, where fluent explanatory output can mask substantive limitations(Lo and Qu, 2025). The remaining, four of five themes captured limitations: complexity barriers, error awareness, insufficient specificity, and active disengagement from the explanatory text.

5.2 Limitations and Implications

Several constraints bound the conclusions that can be drawn. First, the user study recruited 40 participants via convenience sampling, limiting demographic diversity and statistical power; the chart-level analysis in RQ3 is based on only six charts, reducing the inferential weight of what is otherwise the most theoretically significant finding. Second, the qualitative analysis (RQ4) was conducted by a single coder, and seven responses were excluded as off-topic, which may have introduced selection bias into the thematic findings. Third, the error analysis flagged each error type once per explanation, even where the same error appeared multiple times, potentially masking severity or repetition. Finally, both models were evaluated under a single structured prompt; performance under alternative prompting strategies, such as chain-of-thought or retrieval-augmented approaches, remains untested.

Despite these constraints, the findings carry meaningful implications for practice and future research. For healthcare communicators and information designers, the results caution against uncritical deployment of AI-generated chart explanations: surface fluency is not a reliable proxy for factual fidelity, and the comprehension decrement observed here suggests that explanations may displace rather than support active visual engagement. The absence of any protective effect from visualisation experience indicates that miscalibration risk is broadly distributed and unlikely to be resolved by targeting expert audiences. Practically, AI explanation tools in healthcare contexts may need to incorporate explicit uncertainty

signaling, prompted self-verification steps, or accuracy indicators to counteract the over-reliance dynamic identified here. For researchers, the moderate but non-significant RQ3 correlation points to chart-level accuracy as a worthy focus for larger-scale studies: establishing a reliable threshold below which explanation errors measurably harm comprehension would have direct design implications. The HM2 hallucination finding also warrants targeted follow-up; identifying the prompt conditions or chart features that elicit unsupported inferential content could inform both model fine-tuning and deployment guidelines for domain-sensitive applications.

5.3 Conclusions

Taken together, these findings demonstrate a meaningful performance gap between GPT-4o and Claude Sonnet 4.6 in healthcare chart explanation, with unsupported inferential hallucination as Claude's most consistent failure mode. More consequentially for practice, partially accurate AI explanations impaired comprehension while sustaining confidence - a miscalibration pattern that is particularly hazardous in healthcare communication. The absence of any protective effect from visualisation experience suggests the risk of AI-induced over-reliance is broadly distributed. These conclusions reinforce the case for treating explanatory AI not as a transparent communication aid but as an intervention requiring careful calibration-aware design before deployment in high-stakes contexts.

5.4 Personal Reflections

Running this project clarified how early methodological decisions constrain later analytical choices. The small chart-level n in RQ3 reduced the inferential weight of what was theoretically the most interesting finding. Starting again, I would expand the chart pool for the user study, pursue a more diverse participant sample, and formalise inter-rater reliability

from the outset for the qualitative coding - rather than treating single-coder analysis as a post-hoc limitation.

This project reinforced the importance of holding unexpected results with curiosity rather than with anxiety. The finding that AI explanations reduced comprehension initially felt uncomfortable but became the most substantive contribution of the work. I also recognise a tendency to overspecify analytical plans before the data are fully understood. Going forward, I intend to build in structured pause points during analysis to allow emerging patterns to inform the inquiry, rather than treating the pre-registered plan as fixed.

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Appendix

Appendix A

Ethics Form



Downloaded: 19/05/2026
Approved: 09/01/2026

Amina Elshazly
Registration number: 230190252
School of Information, Journalism and Communication
Programme: BSC Data Science

Dear Amina

PROJECT TITLE: Evaluating LLM-Generated Explanations for Healthcare Data Visualisations
APPLICATION: Reference Number 071846

On behalf of the University ethics reviewers who reviewed your project, I am pleased to inform you that on 09/01/2026 the above-named project was **approved** on ethics grounds, on the basis that you will adhere to the following documentation that you submitted for ethics review:

- University research ethics application form 071846 (form submission date: 07/01/2026); (expected project end date: 30/04/2026).
- Participant information sheet 1160576 version 2 (25/12/2025).
- Participant consent form 1160577 version 1 (20/11/2025).

If during the course of the project you need to deviate significantly from the above-approved documentation please inform me since written approval will be required.

Your responsibilities in delivering this research project are set out at the end of this letter.

Yours sincerely

Jared Ahmad
Ethics Admin
School of Information, Journalism and Communication

Please note the following responsibilities of the researcher in delivering the research project:

- The project must abide by the University's Research Ethics Policy: https://www.sheffield.ac.uk/research-services/ethics-integrity/policy
- The project must abide by the University's Good Research & Innovation Practices Policy: https://www.sheffield.ac.uk/polopoly_fs/1.671066!/file/GRIPPolicy.pdf
- The researcher must inform their supervisor (in the case of a student) or Ethics Admin (in the case of a member of staff) of any significant changes to the project or the approved documentation.
- The researcher must comply with the requirements of the law and relevant guidelines relating to security and confidentiality of personal data.
- The researcher is responsible for effectively managing the data collected both during and after the end of the project in line with best practice, and any relevant legislative, regulatory or contractual requirements.

Appendix B

User study Google Form

Group 1 Form

Section 1 of 10

Understanding Healthcare Data Visualizations Study

PARTICIPANT INFORMATION SHEET

Study Title: Evaluating AI Explanations for Healthcare Data Visualizations
Researcher: Amina ElShazly, BSc Data Science, University of Sheffield
Supervsior: Julia Neri, University Teacher in Data Science

What is this study about?
This research examines whether AI-generated explanations help people understand healthcare charts (bar charts, line graphs, scatterplots showing health statistics like vaccination rates or disease trends).

What will I do?
You will complete an online survey (Google Forms), that takes approximately 10-15 minutes to complete where you:

1. View 6 healthcare visualizations (3 with AI explanations, 3 without)
2. Answer 3 multiple-choice comprehension questions per chart
3. Rate your confidence in your answers
4. Provide brief feedback about the explanations

Who can participate?
Adults (18+), English fluent. Participation is voluntary
You can withdraw anytime without penalty.

Is my data confidential?
Yes. Your participation is anonymous:

- No names or emails collected
- Google Forms will be configured to collect anonymous responses and disable automattic email capture
- Data stored securely on university-approved platforms
- Only researcher and supervisor access data
- Data retained one year post-project (June 2026), then deleted
- Complies with GDPR and University of Sheffield ethics policies

Legal basis: Consent for research (GDPR Article 6(1)(a))

Are there risks or benefits?
Risks: Minimal. You'll view publicly available health statistics.
Benefits: Contribute to AI transparency research in healthcare communication.

What happens to results?
Results appear in a BSc dissertation (aggregated, anonymous form only). No individual responses identifiable.

Ethics approval
Approved via University of Sheffield en-bloc ethics route

*

PARTICIPANT CONSENT FORM

Please confirm your agreement by ticking each box:

☐ I confirm that I have read and understood the Participant Information Sheet for the above study. I have h...

- [ ] I understand that my participation is voluntary and that I am free to withdraw at any time during the stud...
- [ ] I understand that after I submit my responses, my data cannot be withdrawn because all responses are c...
- [ ] I understand that my responses will be anonymized, and no personally identifiable information (name, e...
- [ ] I understand that anonymized data from this study may be used in the researcher's dissertation, present...
- [ ] I understand that my anonymized data will be stored securely on password-protected university-approve...
- [ ] I understand that this research has been reviewed and approved through the University of Sheffield's en-b...
- [ ] I understand that the study will take approximately 20 minutes to complete and involves viewing healthc...
- [ ] I confirm that I am 18 years of age or older.
- [ ] I agree to take part in this study.

Title

Data Protection

The University of Sheffield is the Data Controller for this study. Your participation and data will be processed in accordance with data protection legislation, including the UK General Data Protection Regulation (UK GDPR) and Data Protection Act 2018

Contact Information

If you have any questions or concerns about this study, please contact:

Researcher:
Amina ElShazly
Email: aelshazly1@sheffield.ac.uk

For complaints about the conduct of the research, please contact:

Luke Thompson

Head of Data Protection and Legal Services

Email: dataprotection@sheffield.ac.uk

Consent Confirmation

By clicking "Next" below, I confirm that:

I have read the Participant Information Sheet I have ticked all the boxes above I freely consent to participate in this research study

After section 1 Continue to next section

Section 2 of 10

Visualisation literacy

Description (optional)

How often do you view or work with data visualizations (charts, graphs, plots) in your daily life? *

- ○ Never
- ○ Rarely (a few times per year)
- ○ Occasionally (a few times per month)
- ○ Regularly (a few times per week)
- ○ Daily

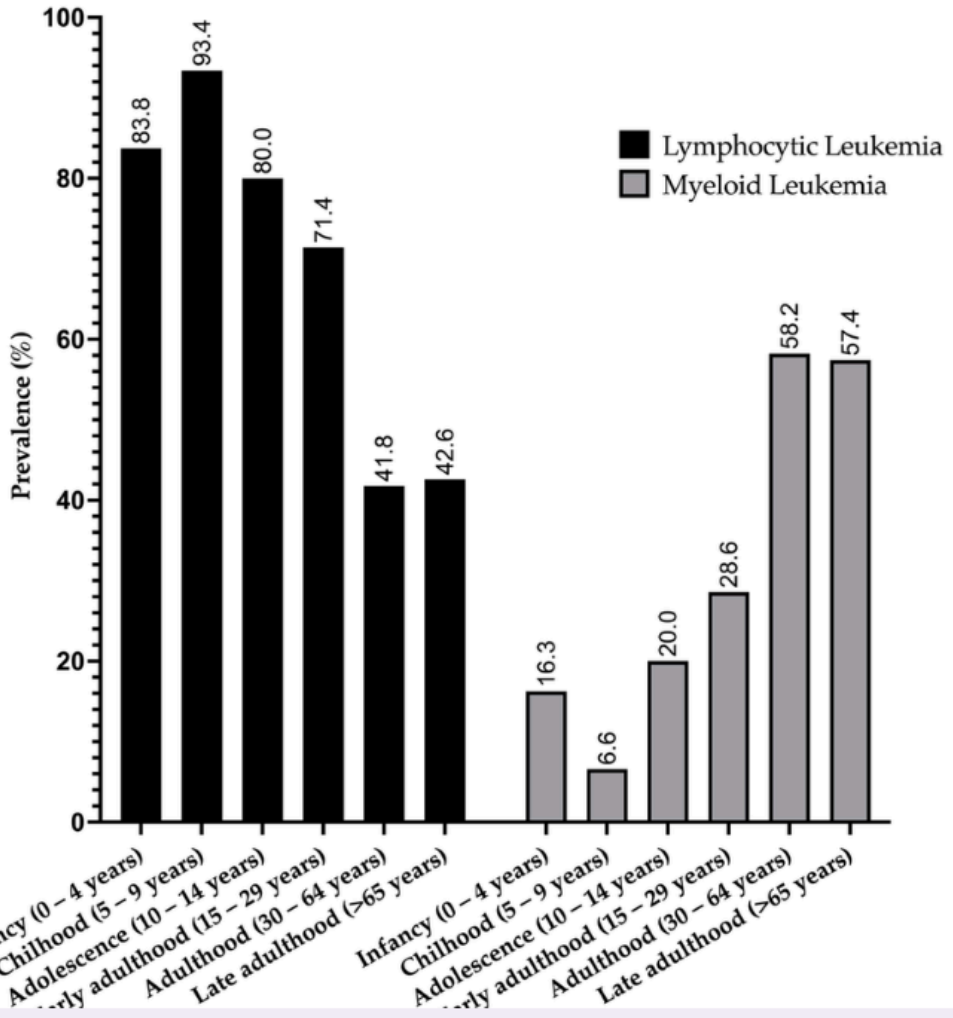
After section 2 Continue to next section

Section 3 of 10

Understanding Healthcare Charts

You will now view 6 healthcare visualizations. For each one, please answer the questions based on what you see in the chart.
Some charts will include a LLM-explanation to help you understand them.

Chart 1: Prevalence of leukemia types in different age groups



Which age group has the highest prevalence of Lymphocytic Leukemia? *

- A) Infancy (0-4 years)
- B) Childhood (5-9 years)
- C) Adolescence (10-14 years)
- D) Early adulthood (15-29 years)

What general trend is shown for Lymphocytic Leukemia prevalence as people age from childhood to late adulthood? *

- A) It steadily increases across all age groups
- B) It first decreases then increases
- C) It steadily decreases across all age groups
- D) It increases to the second age group then decreases

Between the adulthood (30–64 years) and late adulthood (>65 years) age groups, which statement most accurately describes the relationship between the two leukemia types? *

- A) The gap between leukemia types is larger in late adulthood than in adulthood because myeloid leuke...
- B) Myeloid leukemia is dominant in both age groups but its prevalence advantage over lymphocytic leuke...
- C) Lymphocytic leukemia remains dominant in adulthood but myeloid leukemia overtakes it only in late a...
- D) Both leukemia types converge toward equal prevalence in late adulthood suggesting the age-related s...

How confident are you in your answers? *

- 1 - Not at all confident
- 2 - Slightly confident
- 3 - Moderately confident
- 4 - Very confident
- 5 - Extremely confident

After section 3 Continue to next section

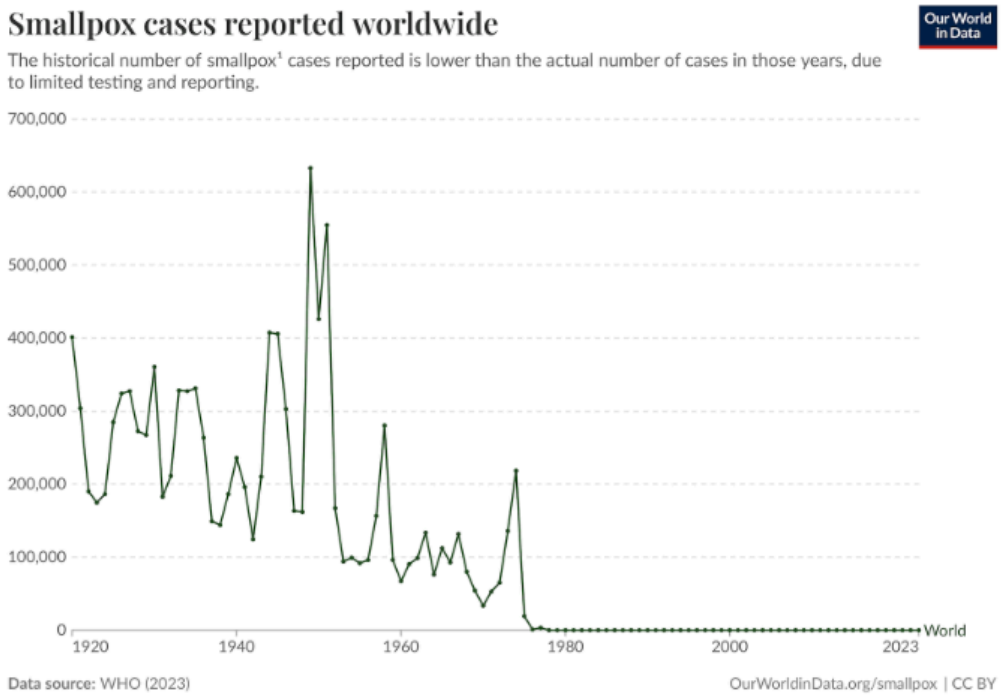
Section 4 of 10

Section title (optional)

Description (optional)

Chart 2: Smallpox cases reported worldwide

This is a line chart showing reported smallpox cases from 1916–2023. Cases remained high and fluctuating throughout the early–mid 20th century before dropping sharply to zero after 1979. The peak reached ~630,000 cases around 1949, while cases fall to 0 from 1980 onward. The most striking comparison is this shift from hundreds of thousands of cases to complete elimination within about 30 years. Overall, a global vaccination campaign successfully eradicated smallpox, marking a major public health achievement.



According to the chart and its caption, in approximately what year was the last reported smallpox case recorded worldwide? *

- A) 1967
- B) 1972
- C) 1980
- D) 1977

What does the chart show about the overall pattern of smallpox cases between 1920 and 1977? *

- A) Cases declined steadily and continuously from 1920 onwards until reaching zero in 1977
- B) Cases remained relatively stable throughout the entire period before dropping sharply to zero after 19...
- C) Cases fluctuated considerably throughout the period with a major peak around the mid-1940s, followe...

D) Cases increased dramatically from 1920 to 1945 then declined steadily with no further fluctuation unti...

Comparing smallpox cases at the 1950 peak to levels recorded in the late 1950s, which statement most accurately describes both the scale and nature of the change? *

A) Cases fell from approximately 630,000 at the 1950 peak to around 300,000 by the late 1950s — a redu...

B) Cases fell from approximately 630,000 at the 1950 peak to around 300,000 by the late 1950s — a redu...

C) Cases fell from approximately 630,000 at the 1950 peak to around 100,000 by the late 1950s — a redu...

D) Cases fell from approximately 630,000 at the 1950 peak to around 300,000 by the late 1950s — a redu...

How confident are you in your answers? *

1 - Not at all confident

2 - Slightly confident

3 - Moderately confident

4 - Very confident

5 - Extremely confident

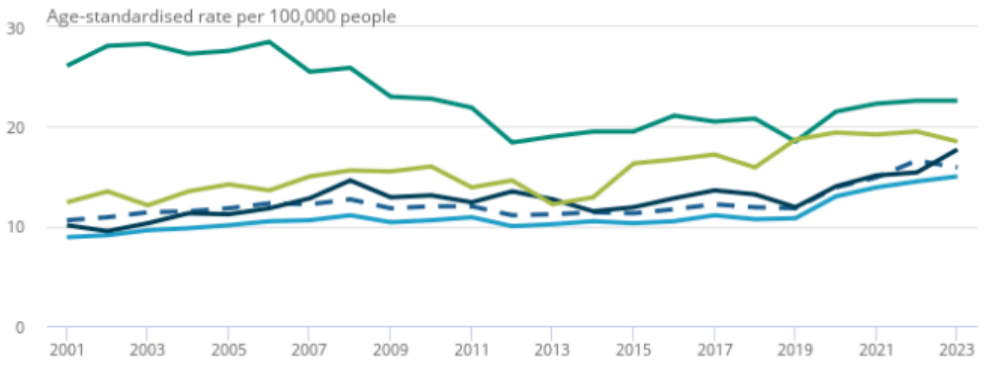
After section 4 Continue to next section

Section 5 of 10

Section title (optional)

Description (optional)

Chart 3: Of the four UK constituent countries, rates of alcohol-specific deaths increased in England and Wales in 2023





What is the lowest rate of mortality per 100, 000 throughout all years? *

- A) 2001-2002
- B) 2001-2008
- C) 2001-2010
- D) 2012-2013

Between 2001 and 2023, which trend best describes Scotland's alcohol-specific death rate? *

- A) Steady increase throughout the period
- B) Remained consistently higher than all other countries with little change
- C) Showed similar patterns to England and Wales throughout
- D) Decreased significantly until 2012, then remained relatively stable

In 2023, how do alcohol-specific death rates in Scotland compare to those in England, and how would you best characterise the size of that difference? *

- A) Scotland's rate is roughly double England's in 2023, representing a dramatic and stark inequality betw...
- B) Scotland's rate in 2023 is approximately 29% higher than England's — a meaningful but not exceptiona...
- C) Scotland and England show virtually identical rates in 2023 having converged steadily since 2011 whe...
- D) Scotland's rate is higher than England's but the gap has widened substantially since 2001 making the ...

How confident are you in your answers? *

- 1 - Not at all confident
- 2 - Slightly confident
- 3 - Moderately confident
- 4 - Very confident

- 3 - Moderately confident
- 4 - Very confident
- 5 - Extremely confident

After section 5 Continue to next section

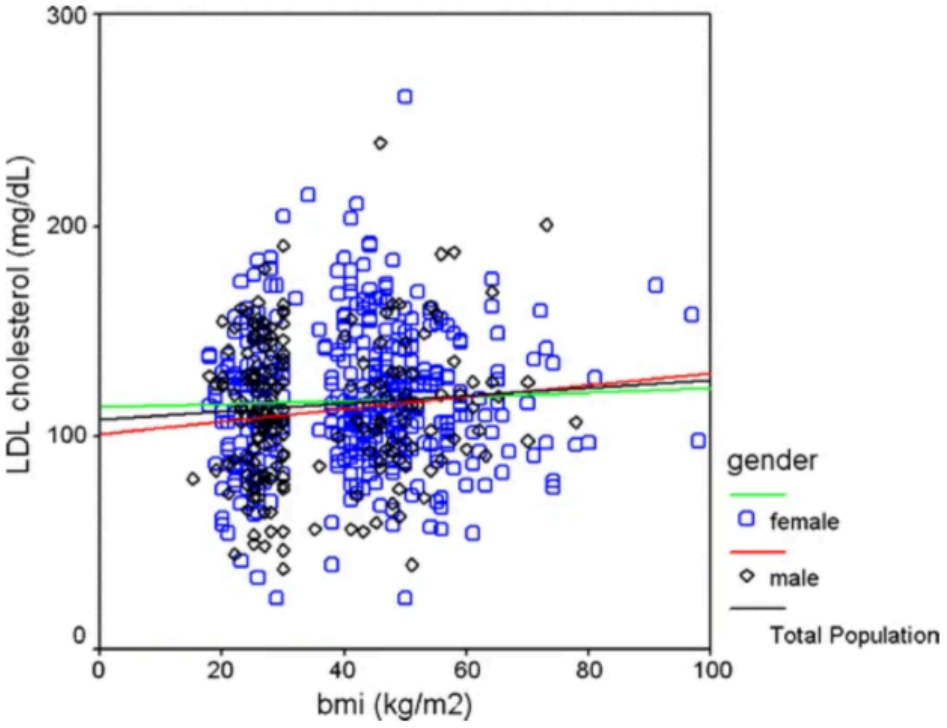
Section 6 of 10

Section title (optional)

Description (optional)

Chart 4: Correlation between LDL cholesterol and BMI, stratified by gender

This is a scatter plot showing the relationship between BMI and LDL cholesterol for males, females, and the overall population. There is a slight upward trend, meaning LDL generally increases modestly as BMI rises, though the relationship is weak. Most individuals cluster between BMI 25–50 kg/m² and LDL 80–140 mg/dL. The highest LDL (~260 mg/dL) occurs at BMI ~45, while the lowest (~20 mg/dL) appears around BMI 30. At higher BMI levels (60–70 kg/m²), male LDL trends slightly higher than female. Overall, BMI alone is not a strong predictor of LDL, and other factors like genetics and lifestyle are important.



According to the chart, approximately where does the majority of data points cluster on the BMI axis? *

- A) BMI 50–200 kg/m²

According to the chart, approximately where does the majority of data points cluster on the BMI axis? *

- A) BMI 50–200 kg/m²
- B) BMI 50–200 mg/dL
- C) BMI 20–60 kg/m²
- D) BMI 25–50 mg/dL

What does the chart show about the overall relationship between BMI and LDL cholesterol levels? *

- A) BMI and LDL cholesterol show a strong positive relationship
- B) There is a slight upward trend showing LDL cholesterol increases modestly as BMI increases
- C) The chart shows no meaningful association between BMI and LDL cholesterol
- D) LDL cholesterol decreases as BMI increases showing an inverse relationship

Which statement most accurately describes the relationship between male and female LDL cholesterol levels across the full BMI range shown in the chart *

- A) Females show consistently higher LDL than males at low BMI values, but males overtake females aro...
- B) Males show higher LDL than females only briefly around BMI 60–70 kg/m² before the lines reconverg...
- C) Females have higher LDL than males at low BMI values, and males only begin to show marginally high...
- D) Males and females show virtually identical LDL levels at all BMI values because both trend lines run p...

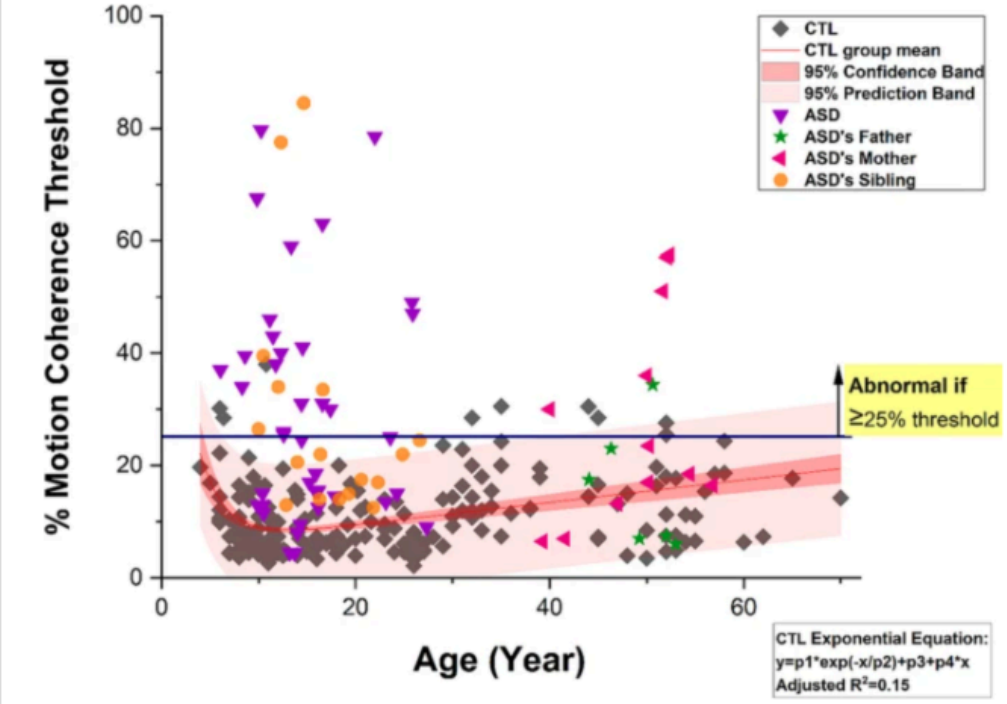
How confident are you in your answers? *

- 1 - Not at all confident
- 2 - Slightly confident
- 3 - Moderately confident
- 4 - Very confident
- 5 - Extremely confident

After section 6 Continue to next section

Chart 5: Percentage thresholds of motion coherence against age of the control individuals and autism family members. CTL = control individuals, ASD = autism spectrum disorder. Total Signal Dots were 1081. The red straight line presents the group mean, 95% confidence interval in dark red band and 95% prediction interval in light red band.

Fig. 1



According to the chart, which group has data points recorded at the highest ages shown, and approximately what is the oldest age visible? *

- A) CTL group, approximately age 70
- B) ASD's sibling, approximately age 85
- C) ASD's Mother group, approximately age 65
- D) CTL group, approximately age 80

What does the chart show about how motion coherence threshold changes with age in the control (CTL) group? *

- A) CTL scores follow a linear downward trend, declining consistently and steadily from childhood throug...
- B) CTL scores improve rapidly during childhood then stabilise, showing motion perception ability increas...
- C) CTL scores show a U-shaped pattern, declining in childhood before rising sharply again in adulthood
- D) CTL scores illustrate a negatively exponential shaped skewed distribution, dropping in early childhood ...

Compared to control participants, how do ASD individuals' motion coherence threshold scores *

Compared to control participants, how do ASD individuals' motion coherence threshold scores differ across all age groups shown in the chart? *

- A) ASD individuals consistently score above 25% across all ages while controls always remain below 25%
- B) ASD individuals generally score higher than controls but both groups show scores above and below th...
- C) ASD individuals and controls show no meaningful difference in scores at any age
- D) Controls score higher than ASD individuals at young ages but lower at older ages

How confident are you in your answers? *

- 1 - Not at all confident
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- 3 - Moderately confident
- 4 - Very confident
- 5 - Extremely confident

After section 7 Continue to next section

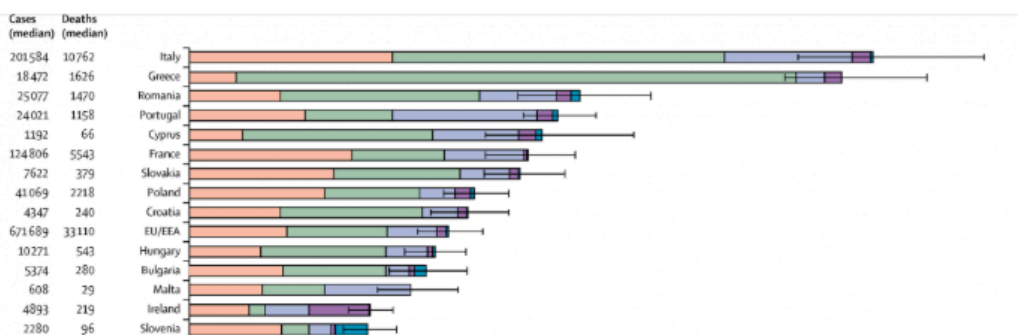
Section 8 of 10

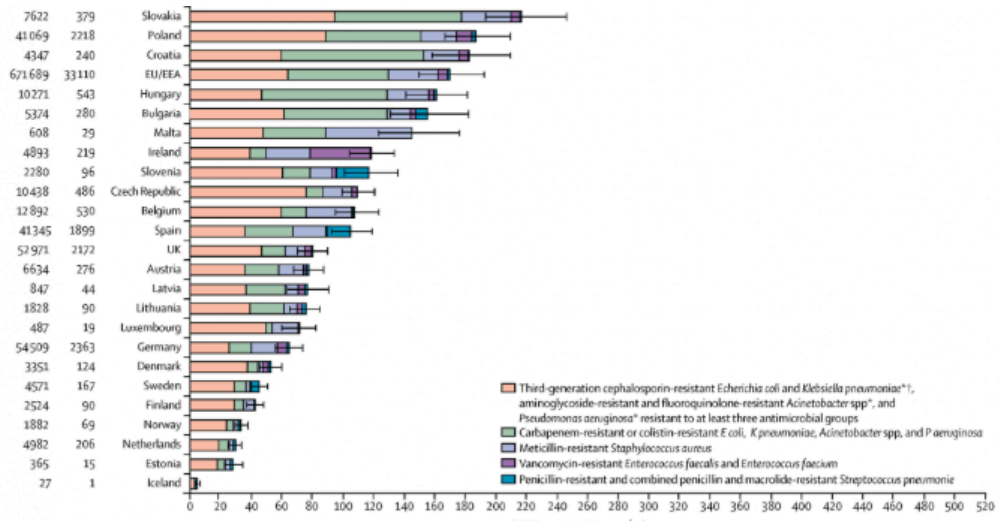
Section title (optional)

Description (optional)

Chart 6: Figure 3 Burden of infections with antibiotic-resistant bacteria in DALYs, EU and European Economic Area, 2015

This is a stacked horizontal bar chart showing the burden of antibiotic-resistant infections across European countries. Southern and Eastern Europe show much higher DALYs than Northern countries. Italy records the highest (~450 DALYs per 100,000), while Iceland is near zero (~1). Italy's burden is about 45 times higher than Iceland's, highlighting a stark disparity. Overall, Southern countries—especially Italy and Greece—face a significantly greater antimicrobial resistance burden, indicating the need for stronger targeted interventions.





According to the chart, which bacterial resistance category contributes the largest portion of Hungary's total DALY burden *

- A) All of them occupy it equally
- B) Third-generation cephalosporin-resistant Echerichia coli and Klebsiella pneumoniae...
- C) Carbapenem-resistant or colistin-resistant E coli, K pneumoniae,...
- D) Meticillin-resistant Staphylococcus aureus (MRSA)

Looking at Ireland's bar composition compared to most other European countries, what is distinctive about Ireland's antibiotic-resistant infection burden? *

- A) The country has the highest DALY burden yet has the second highest burden from Carbapenem-resist...
- B) The country shows a higher proportion of its burden from Meticillin-resistant Staphylococcus aureus (...
- C) The country shows a higher proportion of its burden from Vancomycin-resistant Enterococcus... comp...
- D) The country has an unusually high total DALY burden driven primarily by Carbapenem-resistant organi...

How does Italy's antibiotic-resistant infection burden compare to Iceland's in terms of DALYs per 100,000 population, and which statement most accurately describes the size of that difference? *

- A) Italy's burden is approximately 64 times greater than Iceland's — Italy at around 450 DALYs per 100,00...
- B) Italy's burden is approximately 45 times greater than Iceland's, representing an extreme but predictabl...
- C) Italy's burden is roughly double Iceland's, showing a moderate but meaningful north-south divide in an...

Group 2:

Group 2 contains the same Section 1 and 2 as Group

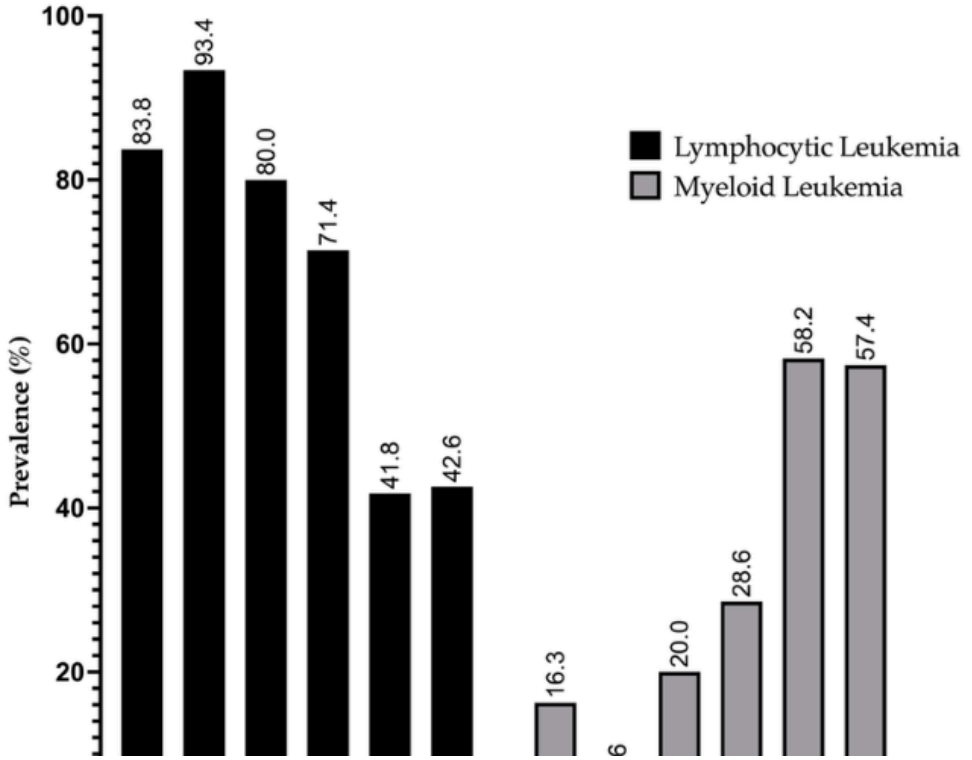
Section 3 of 10

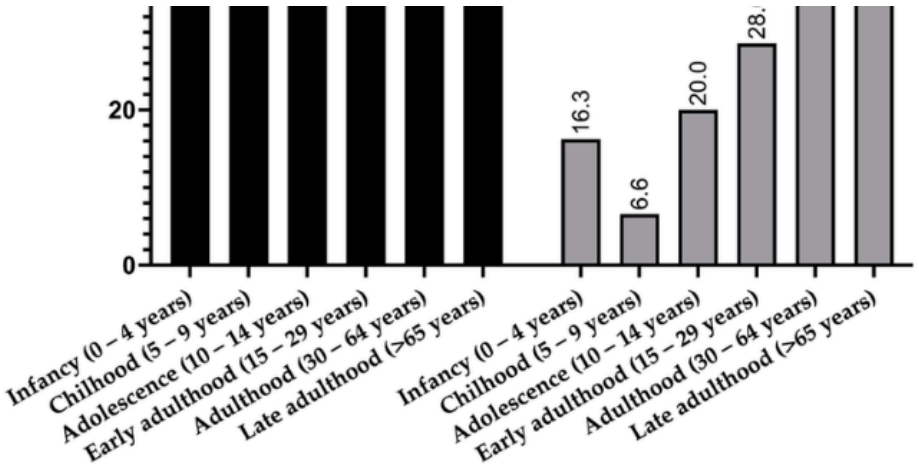
Understanding Healthcare Charts

You will now view 6 healthcare visualizations. For each one, please answer the questions based on what you see in the chart.
Some charts will include a LLM-explanation to help you understand them.

Chart 1: Prevalence of leukemia types in different age groups

A grouped bar chart comparing the prevalence (%) of lymphocytic vs. myeloid leukemia across age groups from infancy to late adulthood. Lymphocytic leukemia is very common in children and teenagers, then declines sharply with age. Myeloid leukemia shows the opposite pattern—rare in children but steadily increases in adults and the elderly. Key takeaway: Different leukemia types dominate at different life stages: lymphocytic in youth, myeloid in older adults.





Which age group has the highest prevalence of Lymphocytic Leukemia? *

- A) Infancy (0-4 years)
- B) Childhood (5-9 years)
- C) Adolescence (10-14 years)
- D) Early adulthood (15-29 years)

What general trend is shown for Lymphocytic Leukemia prevalence as people age from childhood to late adulthood? *

- A) It steadily increases across all age groups
- B) It first decreases then increases
- C) It steadily decreases across all age groups
- D) It increases to the second age group then decreases

Between the adulthood (30–64 years) and late adulthood (>65 years) age groups, which statement most accurately describes the relationship between the two leukemia types? *

- A) The gap between leukemia types is larger in late adulthood than in adulthood because myeloid leuke...
- B) Myeloid leukemia is dominant in both age groups but its prevalence advantage over lymphocytic leuke...
- C) Lymphocytic leukemia remains dominant in adulthood but myeloid leukemia overtakes it only in late a...
- D) Both leukemia types converge toward equal prevalence in late adulthood suggesting the age-related s...

How confident are you in your answers? *

- 1 - Not at all confident
- 2 - Slightly confident

- 1 - Not at all confident
- 2 - Slightly confident
- 3 - Moderately confident
- 4 - Very confident
- 5 - Extremely confident

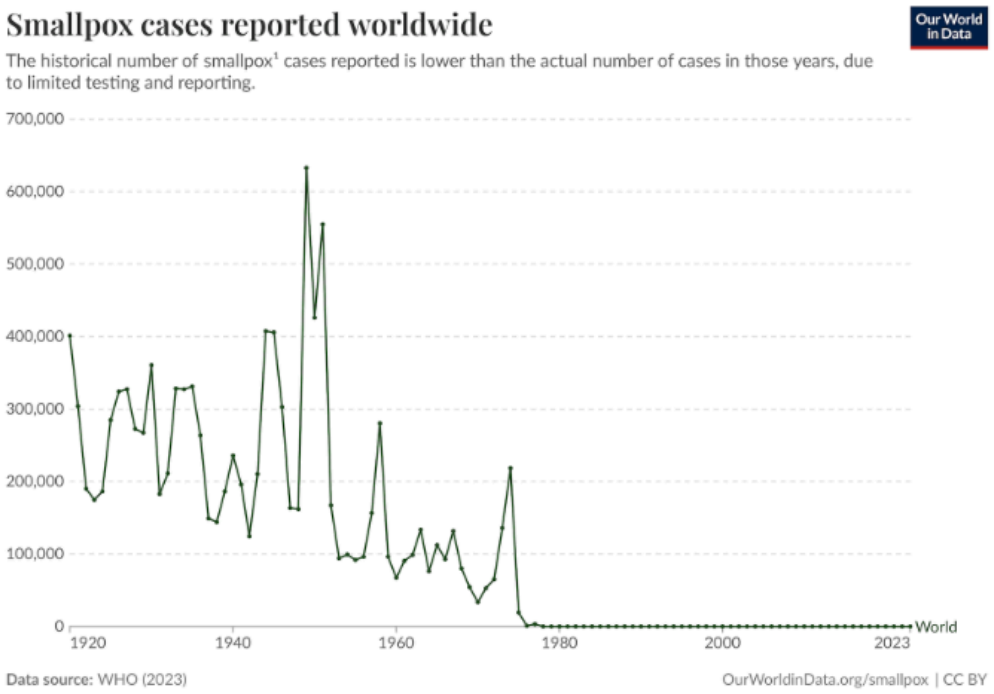
After section 3 Continue to next section

Section 4 of 10

Section title (optional)

Description (optional)

Chart 2: Smallpox cases reported worldwide



According to the chart and its caption, in approximately what year was the last reported smallpox case recorded worldwide? *

- A) 1967
- B) 1972
- C) 1980
- D) 1977

smallpox case recorded worldwide?

- A) 1967
- B) 1972
- C) 1980
- D) 1977

What does the chart show about the overall pattern of smallpox cases between 1920 and 1977? *

- A) Cases declined steadily and continuously from 1920 onwards until reaching zero in 1977
- B) Cases remained relatively stable throughout the entire period before dropping sharply to zero after 19...
- C) Cases fluctuated considerably throughout the period with a major peak around the mid-1940s, followe...
- D) Cases increased dramatically from 1920 to 1945 then declined steadily with no further fluctuation unti...

Comparing smallpox cases at the 1950 peak to levels recorded in the late 1950s, which statement most accurately describes both the scale and nature of the change? *

- A) Cases fell from approximately 630,000 at the 1950 peak to around 300,000 by the late 1950s — a redu...
- B) Cases fell from approximately 630,000 at the 1950 peak to around 300,000 by the late 1950s — a redu...
- C) Cases fell from approximately 630,000 at the 1950 peak to around 100,000 by the late 1950s — a redu...
- D) Cases fell from approximately 630,000 at the 1950 peak to around 300,000 by the late 1950s — a redu...

How confident are you in your answers? *

- 1 - Not at all confident
- 2 - Slightly confident
- 3 - Moderately confident
- 4 - Very confident
- 5 - Extremely confident

After section 4 Continue to next section

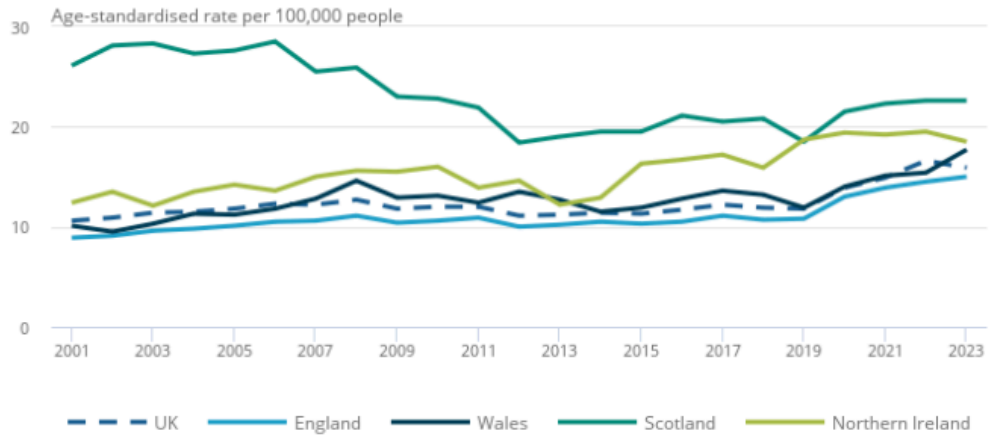
Section 5 of 10

Section title (optional)

Description (optional)

Chart 3: Of the four UK constituent countries, rates of alcohol-specific deaths increased in England and Wales in 2023

This is a multi-line chart showing alcohol-specific death rates across UK nations from 2001–2023. Scotland consistently has the highest rates, while England records the lowest (~9–10 per 100,000 during 2001–2010). All nations show a clear rise after 2019. By 2023, Scotland (~22) remains roughly double England (~17), highlighting persistent geographic inequality. Overall, alcohol-related deaths have increased across the UK, especially post-2019, indicating a growing public health concern despite earlier policy efforts.



What is the lowest rate of mortality per 100, 000 throughout all years? *

- A) 2001-2002
- B) 2001-2008
- C) 2001-2010
- D) 2012-2013

Between 2001 and 2023, which trend best describes Scotland's alcohol-specific death rate? *

- A) Steady increase throughout the period
- B) Remained consistently higher than all other countries with little change
- C) Showed similar patterns to England and Wales throughout
- D) Decreased significantly until 2012, then remained relatively stable

In 2023, how do alcohol-specific death rates in Scotland compare to those in England, and how would you best characterise the size of that difference? *

- A) Scotland's rate is roughly double England's in 2023, representing a dramatic and stark inequality betw...
- B) Scotland's rate in 2023 is approximately 29% higher than England's — a meaningful but not exceptiona...
- C) Scotland and England show virtually identical rates in 2023 having converged steadily since 2011 whe...
- D) Scotland's rate is higher than England's but the gap has widened substantially since 2001 making the ...

How confident are you in your answers? *

- 1 - Not at all confident
- 2 - Slightly confident
- 3 - Moderately confident
- 4 - Very confident
- 5 - Extremely confident

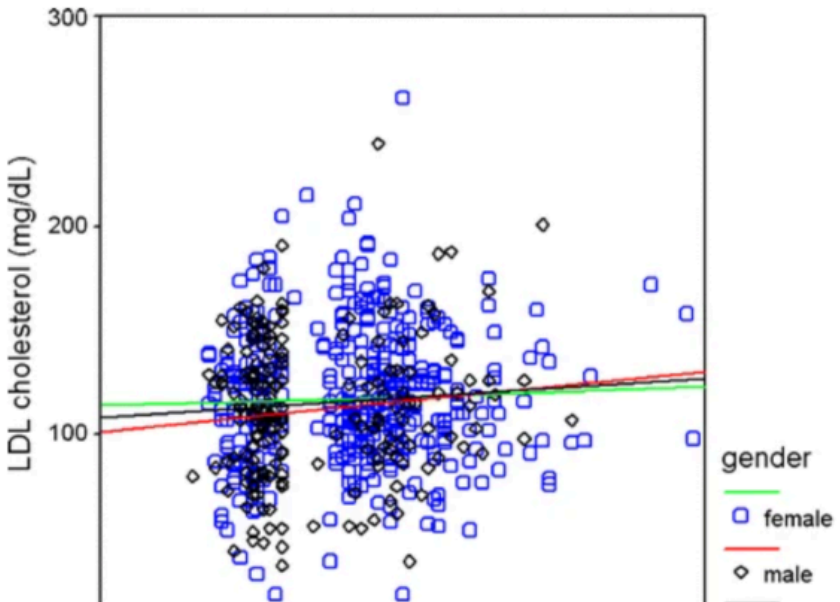
After section 5 Continue to next section

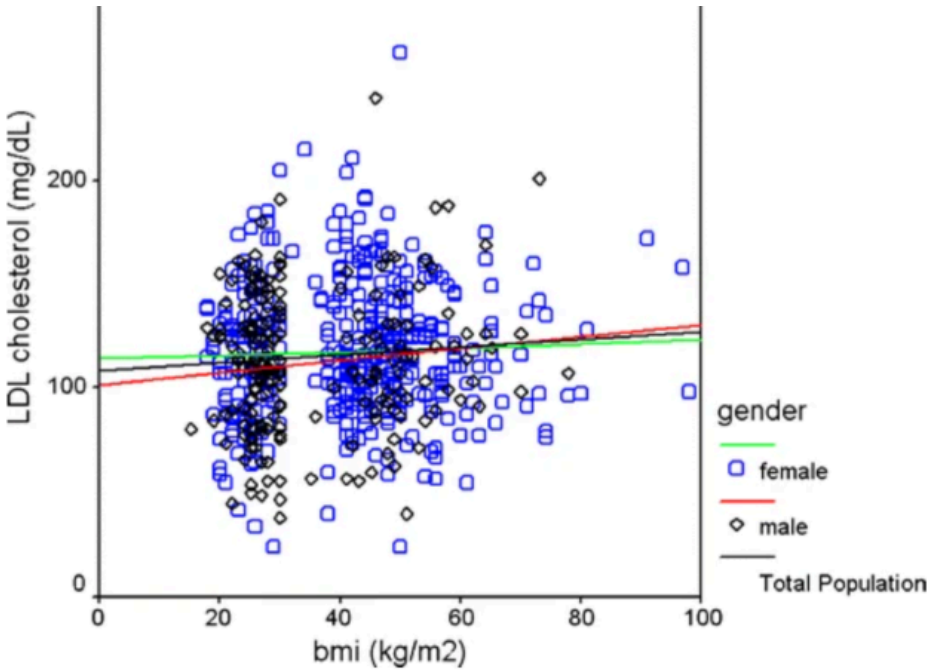
Section 6 of 10

Section title (optional)

Description (optional)

Chart 4: Correlation between LDL cholesterol and BMI, stratified by gender





According to the chart, approximately where does the majority of data points cluster on the BMI axis? *

- A) BMI 50–200 kg/m²
- B) BMI 50–200 mg/dL
- C) BMI 20–60 kg/m²
- D) BMI 25–50 mg/dL

What does the chart show about the overall relationship between BMI and LDL cholesterol levels? *

- A) BMI and LDL cholesterol show a strong positive relationship
- B) There is a slight upward trend showing LDL cholesterol increases modestly as BMI increases
- C) The chart shows no meaningful association between BMI and LDL cholesterol
- D) LDL cholesterol decreases as BMI increases showing an inverse relationship

Which statement most accurately describes the relationship between male and female LDL cholesterol levels across the full BMI range shown in the chart *

- A) Females show consistently higher LDL than males at low BMI values, but males overtake females aro...
- B) Males show higher LDL than females only briefly around BMI 60–70 kg/m² before the lines reconverg...
- C) Females have higher LDL than males at low BMI values, and males only begin to show marginally high...

Which statement most accurately describes the relationship between male and female LDL cholesterol levels across the full BMI range shown in the chart

- A) Females show consistently higher LDL than males at low BMI values, but males overtake females aro...
- B) Males show higher LDL than females only briefly around BMI 60–70 kg/m² before the lines reconverg...
- C) Females have higher LDL than males at low BMI values, and males only begin to show marginally high...
- D) Males and females show virtually identical LDL levels at all BMI values because both trend lines run p...

How confident are you in your answers? *

- 1 - Not at all confident
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After section 6 Continue to next section

Section 7 of 10

Section title (optional)

Description (optional)

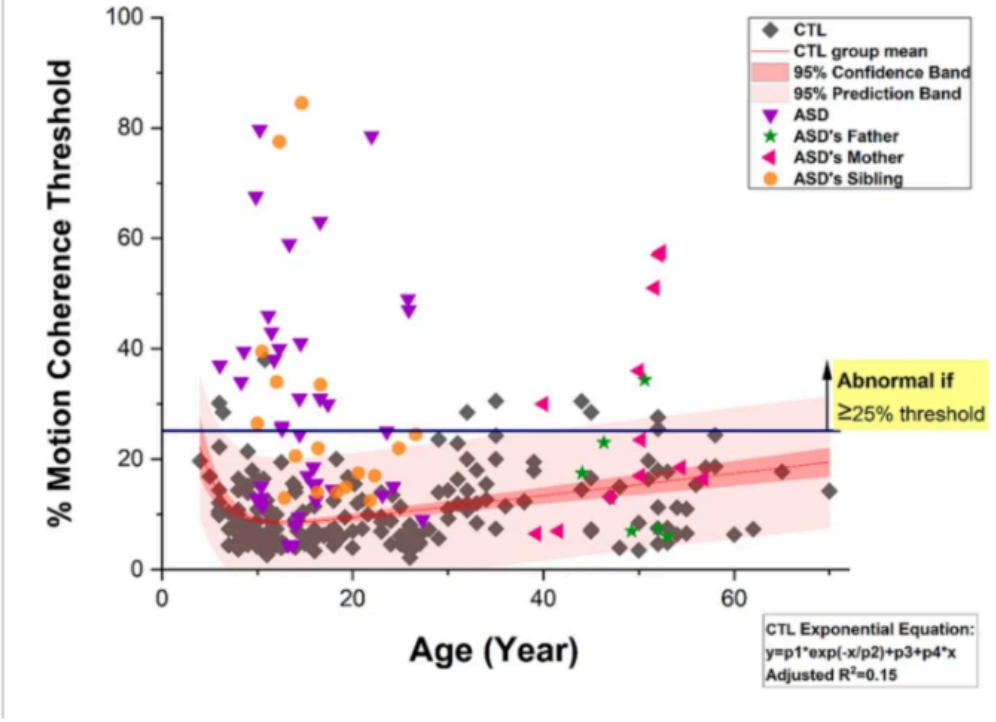
Chart 5: Percentage thresholds of motion coherence against age of the control individuals and autism family members. CTL = control individuals, ASD = autism spectrum disorder. Total Signal Dots were 1081. The red straight line presents the group mean, 95% confidence interval in dark red band and 95% prediction interval in light red band.

This is a multi-group scatter plot showing motion coherence thresholds by age. Controls improve rapidly in childhood and stabilize below the 25% abnormal threshold, while ASD individuals and their family members consistently score above it across all ages. At age ~10, ASD children cluster between 30–80%, whereas controls mostly fall below 25%, highlighting a clear developmental gap. Overall, the 25% threshold separates groups, suggesting motion perception deficits are persistent and may reflect a heritable trait across ASD families rather than only diagnosed individuals.



and may reflect a heritable trait across ASD families rather than only diagnosed individuals.

Fig. 1



According to the chart, which group has data points recorded at the highest ages shown, and approximately what is the oldest age visible? *

- A) CTL group, approximately age 70
- B) ASD's sibling, approximately age 85
- C) ASD's Mother group, approximately age 65
- D) CTL group, approximately age 80

What does the chart show about how motion coherence threshold changes with age in the control (CTL) group? *

- A) CTL scores follow a linear downward trend, declining consistently and steadily from childhood throug...
- B) CTL scores improve rapidly during childhood then stabilise, showing motion perception ability increas...
- C) CTL scores show a U-shaped pattern, declining in childhood before rising sharply again in adulthood
- D) CTL scores illustrate a negatively exponential shaped skewed distribution, dropping in early childhood ...

Compared to control participants, how do ASD individuals' motion coherence threshold scores differ across all age groups shown in the chart? *

- A) ASD individuals consistently score above 25% across all ages while controls always remain below 25%

Compared to control participants, how do ASD individuals' motion coherence threshold scores * differ across all age groups shown in the chart?

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- B) ASD individuals generally score higher than controls but both groups show scores above and below th...
- C) ASD individuals and controls show no meaningful difference in scores at any age
- D) Controls score higher than ASD individuals at young ages but lower at older ages

How confident are you in your answers? *

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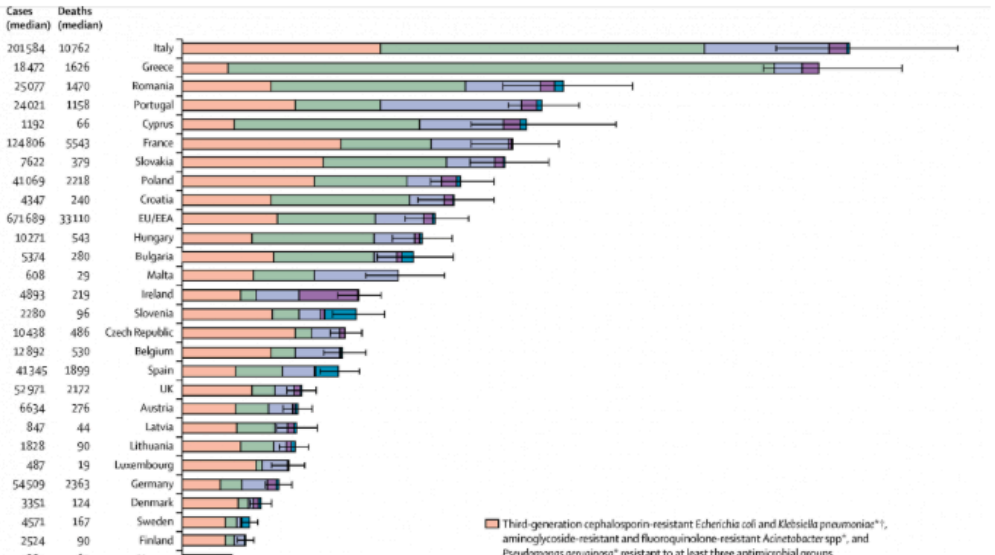
After section 7 Continue to next section

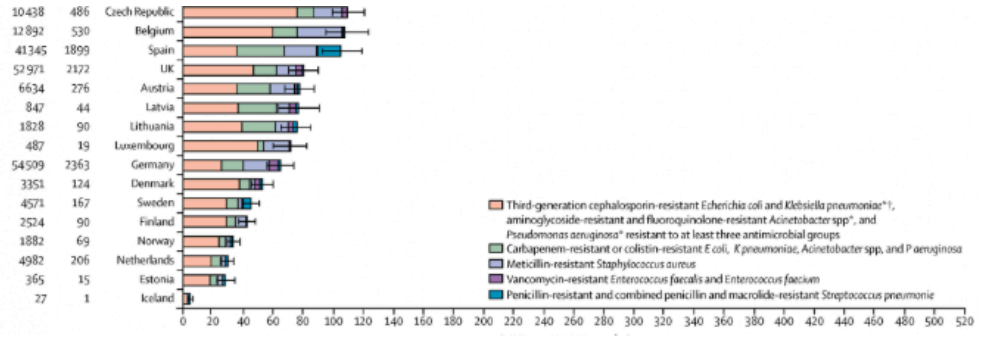
Section 8 of 10

Section title (optional)

Description (optional)

Chart 6: Burden of infections with antibiotic-resistant bacteria in DALYs, EU and European Economic Area, 2015





According to the chart, which bacterial resistance category contributes the largest portion of Hungary's total DALY burden *

- A) All of them occupy it equally
- B) Third-generation cephalosporin-resistant Echerichia coli and Klebsiella pneumoniae...
- C) Carbapenem-resistant or colistin-resistant E coli, K pneumoniae,...
- D) Meticillin-resistant Staphylococcus aureus (MRSA)

Looking at Ireland's bar composition compared to most other European countries, what is distinctive about Ireland's antibiotic-resistant infection burden? *

- A) The country has the highest DALY burden yet has the second highest burden from Carbapenem-resist...
- B) The country shows a higher proportion of its burden from Meticillin-resistant Staphylococcus aureus (...
- C) The country shows a higher proportion of its burden from Vancomycin-resistant Enterococcus... comp...
- D) The country has an unusually high total DALY burden driven primarily by Carbapenem-resistant organi...

How does Italy's antibiotic-resistant infection burden compare to Iceland's in terms of DALYs per 100,000 population, and which statement most accurately describes the size of that difference? *

- A) Italy's burden is approximately 64 times greater than Iceland's — Italy at around 450 DALYs per 100,00...
- B) Italy's burden is approximately 45 times greater than Iceland's, representing an extreme but predictabl...
- C) Italy's burden is roughly double Iceland's, showing a moderate but meaningful north-south divide in an...
- D) Italy and Iceland show similar burdens below 50 DALYs per 100,000 with differences explained entirel...

How confident are you in your answers? *

- 1 - Not at all confident
- 2 - Slightly confident

- 1 - Not at all confident
- 2 - Slightly confident
- 3 - Moderately confident
- 4 - Very confident
- 5 - Extremely confident

After section 8 Continue to next section

Section 9 of 10

Untitled Section

Description (optional)

For the visualizations WITH AI_generated explanations: What made the explanation helpful or unhelpful in answering the questions? Please be specific. *

Long answer text

After section 9 Continue to next section

Section 10 of 10

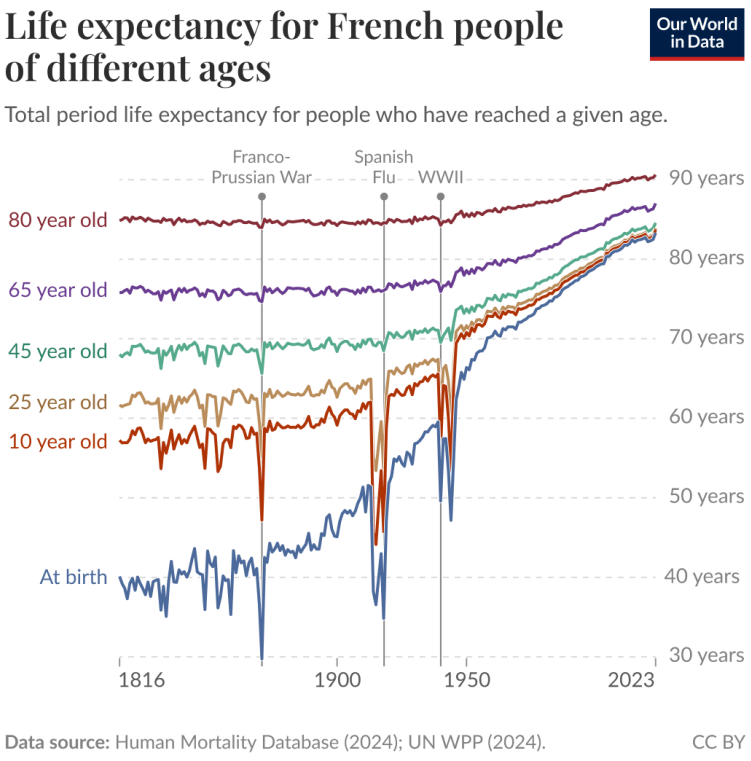
Thank you for completing this survey!

Your responses have been recorded. Your participation helps us understand how to make healthcare information more accessible to everyone.

Appendix C

Table 11

Chart, expert-authored ground truth, and Claude-generated takeaway

Field	Content
Chart	Life expectancy for French people of different ages, 1816–2023 
Source	Our World in Data, period life expectancy by age cohort

Field	Content
Expert-authored caption (ground truth)	“It’s true that child mortality rates were much higher in the past, and their decline has greatly improved overall life expectancy. But in recent decades, improvements in survival at older ages have been even more important. The chart shows the period life expectancy in France for people of different ages. This measures how long someone at each of those ages would live, on average, if they experienced the death rates recorded in that year. For example, the last point on the top dark-red line shows that an 80-year-old in 2023 could expect to live to about 90, assuming mortality rates stayed as they were in 2023. As you can see, life expectancy in France has risen at every age. In 1816, someone who had reached the age of 10 could expect to live to 57. By 2023, this had increased to 84. For those aged 65, it rose from 76 in 1816, to 87 in 2023. “
Claude Sonnet 4.6 takeaway	“1. Chart Type: Multi-line chart tracking six age cohorts over time, with annotated historical events. 2. Axes: X-axis: Year (1816–2023); Y-axis: Total life expectancy in years (30–90 years). 3. Main Trend: Life expectancy improved substantially across all age groups over two centuries, with younger cohorts showing the greatest gains and all groups experiencing sharp dips during major crises.

Field	Content
	4. Key Values: Highest — 80-year-olds reaching ~90 years total life expectancy by 2023; Lowest — life expectancy at birth dropping to approximately 30 years during the Spanish Flu (~1918). 5. Comparison: Life expectancy at birth (~30 years at its 1918 nadir vs. ~83 years by 2023) shows the most dramatic improvement — a near tripling — while the 80-year-old cohort remained relatively stable throughout, improving only modestly from ~82 to ~90 years. 6. Takeaway: This chart reveals that historical gains in French life expectancy were driven predominantly by reducing early-life mortality rather than extending old age. Younger cohorts — especially at birth and age 10 — showed catastrophic drops during the Franco-Prussian War, Spanish Flu, and WWII, yet recovered and climbed dramatically afterward. By 2023, all age groups converge more closely (~78–90 years), reflecting modern medicine's success in tackling infectious disease and childhood mortality, while incremental gains in older cohorts reflect advances in managing age-related conditions.”

Appendix D

Analysis RQ1 in Python

Table 12

Extracting responses from the 2 forms into one dataframe

Python

```
def extract_letter(s):
    """Pulling leading A/B/C/D from strings'."""
    if pd.isna(s):
        return None
    m = re.match(r'^\s*([A-D])\s*\)', str(s))
    return m.group(1) if m else None
def extract_confidence(s):
    """Pull leading digit 1-5"""
    if pd.isna(s):
        return None
    m = re.match(r'^\s*(\d)\s*-', str(s))
    return int(m.group(1)) if m else None


def load_group_responses(filepath, group_label, ai_charts):
    """Returns one row per (participant × chart) for one form."""
    df = pd.read_excel(filepath)
    rows = []
    for pid, r in df.iterrows():
        #column 3 = the frequency-of-use; col 28 = open-ended feedback
        freq_text = r.iloc[3]
                freq_num    =  FREQ_MAP.get(str(freq_text).strip(),  np.nan)  if
pd.notna(freq_text) else np.nan
        open_text = r.iloc[28] if pd.notna(r.iloc[28]) else ''
```

```
        # the 6 chart blocks at columnss 4-27
        for ci in range(1, 7):
            base = 4 + (ci - 1) * 4        #Q1 col
            q1 = extract_letter(r.iloc[base])
            q2 = extract_letter(r.iloc[base + 1])
            q3 = extract_letter(r.iloc[base + 2])
            conf = extract_confidence(r.iloc[base + 3])

            key = ANSWER_KEY[ci]
            q1_correct = int(q1 == key['Q1']) if q1 else 0
            q2_correct = int(q2 == key['Q2']) if q2 else 0
            q3_correct = int(q3 == key['Q3']) if q3 else 0
            chart_score = (q1_correct * Q_WEIGHTS['Q1']
                           + q2_correct * Q_WEIGHTS['Q2']
                           + q3_correct * Q_WEIGHTS['Q3'])

            rows.append({
                'group': group_label,
                'participant_id': f'{group_label}_P{pid + 1:02d}',
                 'frequency_text': str(freq_text).strip() if pd.notna(freq_text)
else None,
                'frequency_num': freq_num,
                'chart': ci,
                'condition': 'AI' if ci in ai_charts else 'no_AI',
                'Q1_answer': q1, 'Q1_correct': q1_correct,
                'Q2_answer': q2, 'Q2_correct': q2_correct,
                'Q3_answer': q3, 'Q3_correct': q3_correct,
                'chart_score': chart_score,
                'confidence': conf,
                'open_response': open_text,
```

```
            })
    return pd.DataFrame(rows)



g1 = load_group_responses(GROUP1_FILE, 'G1', AI_CHARTS_GROUP1)
g2 = load_group_responses(GROUP2_FILE, 'G2', AI_CHARTS_GROUP2)
phase2_long = pd.concat([g1, g2], ignore_index=True)

# Merging chart metadata
phase2_long = phase2_long.merge(CHART_INFO, on='chart', how='left')

n_participants = phase2_long['participant_id'].nunique()
print(f'Total participants: {n_participants} (expected 40)')
print(f'Total response rows: {len(phase2_long)} (expected
{n_participants * 6})')
print()
print('Frequency distribution:')
print(phase2_long.groupby('participant_id').first()['frequency_num']
      .value_counts().sort_index()
            .rename(index={i: f'{i} ({FREQ_LABELS[i]})' for i in
range(5)}))
```

Table 13

Per participant summary table

Variable	Count	Mean	SD	Min	25%	Median	75%	Max
AI Use Frequency	40	2.35	1.19	0	1.00	2.00	3.00	4
Comprehension Score With AI Explanation	40	7.33	3.94	0	4.00	7.00	10.25	16
Comprehension Score Without AI Explanation	40	10.85	4.57	1	7.75	11.00	15.00	18
Total Comprehension Score	40	18.18	7.06	4	13.50	18.50	23.25	32
Total Confidence With AI Explanation	40	10.78	3.10	4	9.00	11.00	13.25	15
Total Confidence Without AI Explanation	40	9.40	3.56	3	7.00	9.50	12.00	15
Mean Confidence With AI Explanation	40	3.59	1.03	1.33	3.00	3.67	4.42	5
Mean Confidence Without AI Explanation	40	3.13	1.19	1.00	2.33	3.17	4.00	5

Appendix E

Statistical analysis conducted on Python

2B. Total accuracy — Wilcoxon signed-rank (paired)

```python
def matched_pair_rb(x, y):
    d = np.array(x) - np.array(y)
    d = d[d != 0]
    if len(d) == 0:
        return 0.0
    abs_ranks = stats.rankdata(np.abs(d))
    R_pos = abs_ranks[d > 0].sum()
    R_neg = abs_ranks[d < 0].sum()
    return (R_pos - R_neg) / (R_pos + R_neg)

gpt_df    = phase1_checklist[phase1_checklist['model'] == 'ChatGPT'].set_index(['chart_type', 'chart_index'])['score']
claude_df = phase1_checklist[phase1_checklist['model'] == 'Claude' ].set_index(['chart_type', 'chart_index'])['score']
paired_charts = pd.DataFrame({'GPT': gpt_df, 'Claude': claude_df}).dropna()
paired_charts['diff'] = paired_charts['GPT'] - paired_charts['Claude']

print(f'Paired charts: {len(paired_charts)}  (expected 45)')
print(f'GPT higher:   {(paired_charts["diff"] > 0).sum()}')
print(f'Claude higher: {(paired_charts["diff"] < 0).sum()}')
print(f'Tied:   {(paired_charts["diff"] == 0).sum()}')
print()

w_stat, w_p = wilcoxon(paired_charts['GPT'], paired_charts['Claude'])
rb_paired = matched_pair_rb(paired_charts['GPT'], paired_charts['Claude'])

print(f'GPT-4o   median = {paired_charts["GPT"].median():.2f}    IQR = [{paired_charts["GPT"].quantile(0.25):.2f}, {paired_charts["GPT"].quantile(0.75):.2f}]')
print(f'Claude   median = {paired_charts["Claude"].median():.2f}    IQR = [{paired_charts["Claude"].quantile(0.25):.2f}, {paired_charts["Claude"].quantile(0.75):
print(f'W = {w_stat:.2f}    p = {w_p:.6g}    {"✓ significant" if w_p < ALPHA else "ns"}')
print(f'matched-pairs rank-biserial r = {rb_paired:+.3f}')
```

```
McNemar per CHECKLIST item with BH-FDR
======================================================================
```

	item	n_pairs	both_1	GPT_only	Claude_only	both_0	discordant	method	p_raw	OR_GPTvsClaude	OR_95CI_lo	OR_95CI_hi	p_bh	sig_bh
0	C1	45	45	0	0	0	0	no discordant	1.0000	NaN	NaN	NaN	1.0000	False
1	C2	45	43	1	1	0	2	exact	1.0000	1.0000	0.1040	9.6139	1.0000	False
2	C3	45	41	0	3	1	3	exact	0.2500	0.1429	0.0074	2.7658	0.5000	False
3	C4	45	32	7	2	4	9	exact	0.1797	3.0000	0.7169	12.5534	0.4492	False
4	C5	45	25	11	1	8	12	exact	0.0063	7.6667	1.3985	42.0301	0.0317	True
5	C6	45	35	4	2	4	6	exact	0.6875	1.8000	0.3836	8.4474	1.0000	False
6	C7	45	14	5	4	22	9	exact	1.0000	1.2222	0.3516	4.2483	1.0000	False
7	C8	45	26	13	3	3	16	exact	0.0213	3.8571	1.1904	12.4982	0.0709	False
8	C9	45	0	23	2	20	25	asymptotic	0.0001	9.4000	2.5519	34.6256	0.0006	True
9	C10	45	20	2	3	20	5	exact	1.0000	0.7143	0.1409	3.6203	1.0000	False

```
Items significant after BH-FDR: 2 / 10
      C5: GPT > Claude  (OR = 7.67, p_BH = 0.0317)
      C9: GPT > Claude  (OR = 9.40, p_BH = 0.0006)
```

```python
# Errors
err_results = pd.DataFrame([mcnemar_paired(c, phase1_errors, ERROR_CODES)
                            for c in ERROR_CODES])
err_results['p_bh'] = multipletests(err_results['p_raw'], method='fdr_bh')[1]
err_results['sig_bh'] = err_results['p_bh'] < ALPHA

print('McNemar per ERROR code with BH-FDR')
print('=' * 72)
display(err_results.round(4))
err_results.to_csv(f'{TBL_DIR}/rq1_mcnemar_errors.csv', index=False)
print(f'\nError codes significant after BH-FDR: {err_results["sig_bh"].sum()} / 9')
if err_results['sig_bh'].any():
    for _, r in err_results[err_results['sig_bh']].iterrows():
        direction = 'GPT more frequent' if r['GPT_only'] > r['Claude_only'] else 'Claude more frequent'
        print(f'    {r["item"]:>5}: {direction}  (OR = {r["OR_GPTvsClaude"]:.2f}, p_BH = {r["p_bh"]:.4f})')
```

```
McNemar per ERROR code with BH-FDR
========================================================================
```

	item	n_pairs	both_1	GPT_only	Claude_only	both_0	discordant	method	p_raw	OR_GPTvsClaude	OR_95CI_lo	OR_95CI_hi	p_bh	sig_bh
0	FE1	45	12	4	12	17	16	exact	0.0768	0.3600	0.1226	1.0574	0.2304	False
1	FE2	45	1	4	5	35	9	exact	1.0000	0.8182	0.2354	2.8439	1.0000	False
2	FE3a	45	6	2	2	35	4	exact	1.0000	1.0000	0.1732	5.7723	1.0000	False
3	FE3b	45	22	4	4	15	8	exact	1.0000	1.0000	0.2707	3.6938	1.0000	False
4	FE4	45	5	7	7	26	14	exact	1.0000	1.0000	0.3634	2.7515	1.0000	False
5	FE5	45	12	1	11	21	12	exact	0.0063	0.1304	0.0238	0.7151	0.0286	True
6	HM1	45	0	2	3	40	5	exact	1.0000	0.7143	0.1409	3.6203	1.0000	False
7	HM2	45	20	2	23	0	25	asymptotic	0.0001	0.1064	0.0289	0.3919	0.0006	True
8	OE1	45	19	4	2	20	6	exact	0.6875	1.8000	0.3836	8.4474	1.0000	False

```
Error codes significant after BH-FDR: 2 / 9
    FE5: Claude more frequent  (OR = 0.13, p_BH = 0.0286)
    HM2: Claude more frequent  (OR = 0.11, p_BH = 0.0006)
```

2F. Accuracy by chart type — Kruskal-Wallis + Dunn's post-hoc

```python
def eta_squared_kw(h, n, k):
    return max(0.0, (h - k + 1) / (n - k))

def kw_with_dunn(df, group_col, value_col):
    groups = [g[value_col].values for _, g in df.groupby(group_col, observed=True) if len(g) > 0]
    names  = [n for n, g in df.groupby(group_col, observed=True) if len(g) > 0]
    H, p = kruskal(*groups)
    n_total = sum(len(g) for g in groups)
    eps2 = eta_squared_kw(H, n_total, len(groups))
    return H, p, eps2, names, groups

H, p, eps2, names, groups = kw_with_dunn(phase1_checklist, 'chart_type', 'score')
print('Kruskal-Wallis: checklist score by chart type (pooled across models)')
print('-' * 72)
print(f'Groups: {names}')
print(f'H = {H:.3f}    p = {p:.4f}    ε² = {eps2:.3f}    {"✓" if p < ALPHA else "ns"}')
print('\nPairwise Mann-Whitney (Bonferroni corrected):')
pairs = [(0, 1), (0, 2), (1, 2)]
for i, j in pairs:
    u, pp = mannwhitneyu(groups[i], groups[j], alternative='two-sided')
    pp_corr = min(1.0, pp * len(pairs))
    print(f'  {names[i]:<8} vs {names[j]:<8}  U = {u:.1f}  p = {pp:.4f}  p_corr = {pp_corr:.4f}')
```

```
Kruskal-Wallis: checklist score by chart type (pooled across models)
------------------------------------------------------------------------
Groups: ['bar', 'line', 'scatter']
H = 3.592    p = 0.1660    ε² = 0.018    ns

Pairwise Mann-Whitney (Bonferroni corrected):
  bar      vs line      U = 495.0  p = 0.5018  p_corr = 1.0000
  bar      vs scatter   U = 568.5  p = 0.0768  p_corr = 0.2305
  line     vs scatter   U = 538.5  p = 0.1876  p_corr = 0.5629
```

```
def safe_chi2(tab):
    arr = tab.values
    if arr.sum() == 0 or arr.shape[0] < 2 or arr.shape[1] < 2:
        return {'test': 'n/a', 'stat': np.nan, 'p': np.nan, 'V': np.nan}
    chi2, chi_p, _, expected = chi2_contingency(arr)
    if arr.shape == (2, 2) and (expected < 5).any():
        _, fp = fisher_exact(arr)
        n = arr.sum()
        return {'test': 'Fisher', 'stat': chi2, 'p': fp, 'V': np.sqrt(chi2 / n)}
    n = arr.sum()
    r, c = arr.shape
    V = np.sqrt(chi2 / (n * (min(r, c) - 1)))
    return {'test': 'Chi2', 'stat': chi2, 'p': chi_p, 'V': V}
rows = []
for c in ERROR_CODES:
    tab_t = pd.crosstab(phase1_errors[c].astype(int), phase1_errors['chart_type'])
    r = safe_chi2(tab_t); r['error'] = c; r['factor'] = 'chart_type'; rows.append(r)
    tab_c = pd.crosstab(phase1_errors[c].astype(int), phase1_errors['complexity'])
    r = safe_chi2(tab_c); r['error'] = c; r['factor'] = 'complexity'; rows.append(r)

err_dist = pd.DataFrame(rows)[['error', 'factor', 'test', 'stat', 'p', 'V']]
print("Chi-square / Fisher for error code × factor, with Cramér's V")
display(err_dist.round(4))
err_dist.to_csv(f'{TBL_DIR}/rq1_error_dist.csv', index=False)
```

```
Chi-square / Fisher for error code × factor, with Cramér's V
```

```
def false_confidence_count(g):
    out = {}
    for cond in ['AI', 'no_AI']:
        sub = g[g['condition'] == cond]
        hcw = ((sub['confidence'] >= 4) & (sub['chart_score'] < MAX_PER_CHART)).sum()
        out[f'false_conf_{cond}'] = int(hcw)
    return pd.Series(out)


fc = (phase2_long.groupby('participant_id')
      .apply(false_confidence_count, include_groups=False).reset_index())
participants = participants.merge(fc, on='participant_id', how='left')

w_fc, p_fc = wilcoxon(participants['false_conf_AI'], participants['false_conf_no_AI'])
rb_fc = matched_pair_rb(participants['false_conf_AI'], participants['false_conf_no_AI'])
print('Wilcoxon signed-rank: false-confidence count AI vs no-AI (0–3 each)')
print('-' * 72)
print(f'Mean AI    = {participants["false_conf_AI"].mean():.2f}')
print(f'Mean no-AI = {participants["false_conf_no_AI"].mean():.2f}')
print(f'W = {w_fc:.2f}    p = {p_fc:.6g}    {"✓" if p_fc < ALPHA else "ns"}')
print(f'matched-pairs rank-biserial r = {rb_fc:+.3f}')

m_ai = participants['false_conf_AI'].mean()
m_no = participants['false_conf_no_AI'].mean()
if m_no > 0:
    print(f'\nFalse-confidence ratio (AI / no-AI) = {m_ai / m_no:.2f}×')
```

```
ilcoxon signed-rank: false-confidence count AI vs no-AI (0–3 each)
```

```python
# error-question table
error_rows = []
for pid, gp in phase2_long.groupby('participant_id'):
    for (chart, qkey), distractor in ERROR_DISTRACTORS.items():
        row = gp[gp['chart'] == chart]
        if row.empty: continue
        r = row.iloc[0]
        if r['condition'] != 'AI': continue
        given = r[f'{qkey}_answer']
        if given is None: continue
        correct = ANSWER_KEY[chart][qkey]
        if given == correct: status = 'resisted'
        elif given == distractor: status = 'aligned'
        else: status = 'other'
        error_rows.append({'participant_id': pid, 'frequency_num': gp['frequency_num'].iloc[0],
                           'chart': chart, 'question': qkey, 'given': given,
                           'correct': correct, 'distractor': distractor,
                           'status': status, 'confidence': r['confidence']})
error_responses = pd.DataFrame(error_rows)

```

7. RQ4 — Chart-level accuracy vs user outcomes (descriptive, n=6)

```python
ai_only = phase2_long[phase2_long['condition'] == 'AI'].copy()
chart_lvl = (ai_only.groupby('chart')
             .agg(mean_comp=('chart_score', 'mean'),
                  median_comp=('chart_score', 'median'),
                  mean_conf=('confidence', 'mean'),
                  median_conf=('confidence', 'median'),
                  n=('chart_score', 'count')).reset_index()
             .merge(CHART_INFO, on='chart'))

print('── RQ4 CHART-LEVEL CHECK ──')
print(f'Charts summarised: {len(chart_lvl)} (expected 6)')
display(chart_lvl)
chart_lvl.to_csv(f'{TBL_DIR}/rq4_chart_level.csv', index=False)

rho_b1, p_b1 = spearmanr(chart_lvl['accuracy'], chart_lvl['mean_comp'])
rho_b2, p_b2 = spearmanr(chart_lvl['accuracy'], chart_lvl['mean_conf'])

print('\nRQ4 — Spearman ρ across the 6 user-study charts (descriptive, n=6)')
print(f'Accuracy × mean comprehension: ρ = {rho_b1:+.3f}, p = {p_b1:.4f}')
print(f'Accuracy × mean confidence:    ρ = {rho_b2:+.3f}, p = {p_b2:.4f}')
```

```
── RQ4 CHART-LEVEL CHECK ──
Charts summarised: 6 (expected 6)
```

	chart	mean_comp	median_comp	mean_conf	median_conf	n	type	complexity	accuracy	has_error
0	1	3.4000	4.0000	4.1500	4.0000	20	bar	moderate	8	False
1	2	2.8500	3.0000	3.3000	3.0000	20	line	simple	9	False
2	3	3.0000	3.0000	4.0000	4.0000	20	line	moderate	7	True
3	4	0.9500	0.0000	3.7000	4.0000	20	scatter	moderate	4	True
4	5	2.0000	1.0000	3.1500	3.5000	20	scatter	complex	5	True
5	6	2.4500	2.0000	3.2500	3.0000	20	bar	complex	6	True

Appendix F

Table 1. Checklist coding rules (C1–C10)

Code	Checklist item	When it flags (fails)
C1	Chart type identified	Flags when the explanation misidentifies the chart type or fails to specify a fundamental structural element such as the number of panels, the stacking convention, or whether the chart is faceted.
C2	X-axis identified	Flags when the x-axis label, range, or units are missing or wrong, including failing to name the axis variable correctly, stating the wrong range, or omitting unit qualifiers.
C3	Y-axis identified	Flags when the y-axis label, range, units, or scale type (linear/log) are missing or wrong, including misidentifying a log scale as linear or omitting unit qualifiers.
C4	Highest value correct	Flags when the highest value reported is numerically wrong or attributed to the wrong entity.
C5	Lowest value correct	Flags when the lowest value reported is numerically wrong or attributed to the wrong entity. Also flags when the extremum is incomplete, naming only one entity when several share the minimum value.
C6	Direction correct	Flags when the explanation asserts the wrong direction of a trend, denies a visible direction, or asserts a direction across entities that did not all share it.
C7	Pattern correct	Flags when the pattern shape, strength, or coverage descriptor is wrong, or when the paper's specified pattern type is missed.
C8	Comparison correct	Flags when a comparison between two entities, periods, or groups uses wrong values, wrong ratios, wrong magnitude descriptors, or attributes the wrong direction to one side of the comparison.
C9	Takeaway free of fabrication	Flags when the takeaway introduces information not present in the chart, including fabricated factual content or causal, policy, or clinical interpretations not supported by the chart's literal data.

C10	Captures paper's primary finding	Flags when the takeaway misses or contradicts the paper's central analytical claim, including missing paper-specified thresholds, missing counter-intuitive findings the paper emphasises, or directly inverting the paper's primary point.

Table 2. Error taxonomy coding rules

Code	Error type	When it fires
FE1	Wrong value	Fires when a numerical claim is wrong, including wrong values at extremums, wrong proportions or percentages, wrong year for a peak, wrong ratio claims, wrong threshold values, or counts of categories that do not match the chart. A tolerance of approximately 5% is allowed before firing.
FE2	Axis/label error	Fires when an axis label, axis range, unit, category name, colour-to-category mapping, or count of chart elements is wrong.
FE3a	Wrong direction	Fires when the direction of a trend is wrong or denied, including claiming positive when negative, stable when rising, no relationship when correlation is visible, or asserting a direction for entities that did not change direction. Distinct from FE3b, which addresses pattern shape rather than direction.
FE3b	Wrong pattern characterisation	Fires when a pattern shape, strength, or coverage descriptor mismatches the chart or the paper's specified pattern, including misnaming the pattern type, overstating coverage, or oversimplifying ordering. Distinct from FE4, which addresses intensity rather than shape.
FE4	Wrong magnitude descriptor	Fires when a magnitude or intensity descriptor mismatches the actual change, including understating substantial change, overstating minor differences, mismatching paper-specified magnitudes, or applying ratio claims that do not match the actual ratio.
FE5	Wrong entity at extremum	Fires when the entity named at the highest or lowest value is wrong, when the right value is attached to the wrong entity (right magnitude, wrong period or group), or when the extremum group is incomplete.

HM1	Fabricated factual content	Fires when the explanation introduces specific factual content not present in the chart, including fabricated substances, organisations, time periods, locations, populations, or qualifiers. Distinct from HM2, which addresses causal layering rather than invented factual content.
HM2	Causal/policy/clinical mechanism, hallucinated	Fires when the explanation introduces causal interpretations, policy prescriptions, clinical recommendations, or external mechanisms not present in the chart.
OE1	Omission of paper's primary finding	Fires when the takeaway misses or contradicts the paper's central analytical claim, including missing strength or significance characterisations the paper specifies, missing counter-intuitive findings the paper emphasises, missing specific statistical thresholds, or directly inverting the paper's primary framing.

Appendix G

Bar charts:

- Simple: 1 data series, fewer than 12 bars, single colour, no error bars
- Moderate: 2–5 data series, multiple colours, legend, optional error bars
- Complex: 5+ data series, multiple colours and patterns, error bars, annotations

Line graphs:

- Simple: 1 data series, single colour
- Moderate: 2–6 data series, multiple colours
- Complex: 6+ data series, overlapping lines, confidence bands or dual y-axes or faceted, multiple colors

Scatter plots:

- Simple: 1 data series, single colour, optional single trend line
- Moderate: up to 2 data series, 2 colour-coded groups maximum, trend lines, optional size encoding
- Complex: 2+ data series, multiple encodings (colour, size, shape), multiple trend lines, confidence bands or faceted